VayuNetra

AI-Powered Urban Air Quality Intelligence for Smart-City Intervention

"We don't just measure the air. We trace it, predict it, and act on it."

Document	**Product Requirements Document + Technical Architecture (Combined)**
Problem Statement	**PS5 — AI-Powered Urban Air Quality Intelligence**
Hackathon	**Economic Times AI Hackathon 2026 (2nd Edition)**
Goal	**Win #1**
Version	**v1.4 (PRD & Architecture in sync)**
Date	**26 June 2026**

Product Requirements Document (PRD)

VayuNetra — AI-Powered Urban Air Quality Intelligence for Smart-City Intervention

> **Tagline:** *"We don't just measure the air. We trace it, predict it, and act on it."* **Positioning wedge:** Every existing tool **measures** pollution. VayuNetra **assigns the source, forecasts the spike, and triggers the intervention.** We are an *action engine*, not another dashboard.

Document Control

Field	Value
Project / Product name	VayuNetra ("the eye on the air") — *working name; alternates: VayuDrishti, AirIQ, SourceAQ, PranaGrid*
Hackathon	Economic Times AI Hackathon 2026 (2nd Edition)
Problem Statement	**PS5 — AI-Powered Urban Air Quality Intelligence for Smart City Intervention**
Theme	Smart Cities / Environmental Intelligence / Geospatial Analytics / Public Health
Goal	Win **#1**. Nothing below.
PRD version	v1.4 (synced with ARCHITECTURE.md)
Status	Draft for team alignment
Team size	2–3 people (lean, full-stack capable)
Build timeline	Time is **not** a constraint — phase-gated, ambition-maximised
Infra cost	**₹0 — 100% free-tier** (no paid services anywhere; see ARCHITECTURE.md §5)
Target cities	**City-agnostic architecture from day one**; showcase trio: **Delhi** (depth + impact), **Bengaluru** (Kannada advisory + "clean city deteriorating" story), **Mumbai** (coastal dispersion + Marathi)

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1. Executive Summary

India's air-quality crisis kills an estimated **1.67 million people every year** (Lancet Planetary Health) and is no longer a Delhi-only problem — it is a national urban emergency. The country has already deployed **900+ Continuous Ambient Air Quality Monitoring Stations (CAAQMS)**, yet a **2024 CAG audit found only 31%** of cities with monitoring data had *any* actionable multi-agency response protocol linked to those readings. **The data exists. The intelligence layer to act on it does not.**

City administrators do not need another dashboard that shows them a red number. They need three things that **do not exist together anywhere today**:

1. **Source attribution** — *which* sources are responsible at *this* location, *right now*.
2. **Hyperlocal forecasting** — what AQI will be in 24–72 hours at **ward / 1 km** resolution.
3. **Enforcement intelligence** — *where* to deploy limited inspectors for maximum impact.

VayuNetra is a city-agnostic, multi-agent AI platform that fuses CAAQMS ground sensors, **Sentinel-5P / MODIS satellite** data, mobility feeds, meteorological forecasts, and geospatial land-use layers into a single **action engine**. It runs five coordinated AI agents — **Source Attribution, Hyperlocal Forecast, Enforcement Prioritisation, Citizen Advisory, and Multi-City Comparative Intelligence** — that move a city from *reactive monitoring* to *proactive, evidence-based intervention*.

Our North-Star Metric: collapse the **time from pollution signal → validated, source-attributed enforcement action** from the *weeks-or-never* the CAG audit documents, down to **minutes**.

Why we win #1: we hit every judging axis deliberately — a genuinely novel **satellite-ground "blame map"** (Innovation), a forecast that **provably beats the persistence baseline** with numbers (Technical Excellence + Business Impact), a **city-agnostic engine demoed across 3 cities** (Scalability), and a **map-first console + multi-language citizen advisory** (User Experience) — all wrapped in an India-scale, life-and-death business case.

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2. Problem Context (faithful to PS5)

> This section preserves the exact framing and figures from the official PS5 brief so the whole team works from ground truth.

The crisis is national, not local.

- **Delhi** averaged an **AQI of 218** (classified *Poor* or worse) for **over 200 days** in 2024-25.
- **Mumbai** recorded dangerous AQI levels on **over 60 days** in 2024.
- **Kolkata** averaged AQI **above 150** for large parts of the winter season.
- **Bengaluru and Chennai** — long considered relatively clean — have seen **measurable deterioration** as vehicle density and construction activity surge.
- CPCB's National Air Quality data for 2024 shows **24 of India's 50 most polluted cities are Tier-1 or Tier-2 urban centres**.

The human cost. The Lancet Planetary Health journal estimated **1.67 million premature deaths annually** from air pollution in India — a public-health burden falling disproportionately on urban populations.

The intelligence gap. Despite **900+ CAAQMS** under the National Clean Air Programme (NCAP), a **2024 CAG audit found only 31%** of cities with monitoring data had any actionable multi-agency response protocols linked to those readings.

The official challenge (verbatim intent). *Build an AI-powered Urban Air Quality Intelligence platform that fuses monitoring station data, satellite imagery, mobility feeds, meteorological forecasts, and geospatial land-use layers to move from reactive monitoring to proactive, evidence-based intervention — giving city administrators the tools to reduce pollution **at source** rather than just measure it.*

What the brief explicitly says cities need (our design contract):

> *"City administrations need more than dashboards. They need geospatial attribution (which sources are responsible at this location, right now), predictive forecasting (what will AQI be in 24 hours at ward level), and enforcement intelligence (where to deploy inspectors for maximum impact). That combination does not exist today."*

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3. The Gap & Our Thesis

3.1 Why existing solutions fall short

Existing class	What it does	What it *can't* do
CPCB / SAFAR dashboards	Show current & historical AQI	No source attribution, no hyperlocal forecast, no enforcement targeting
Consumer apps (IQAir, etc.)	Display AQI + generic "wear a mask" advice	No authority workflow, no source-level action, no India-language local advisory
Academic apportionment studies	Accurate source split	One-off, months-late, not real-time, not operational
GRAP (Graded Response Action Plan)	Rule-based escalation	Coarse (city-wide), reactive, not predictive, not source-targeted

The white space: nobody fuses **real-time** attribution + **hyperlocal** forecast + **operational** enforcement + **citizen** advisory in one continuously-updating, city-agnostic system.

3.2 Our thesis

> **Pollution is not an information problem. It is an *action* problem.** The signal is everywhere (900+ stations, free satellite passes, open weather). What's missing is the layer that turns signal → **attributed cause** → **predicted spike** → **prioritised, evidence-backed action** → **protected citizen**. VayuNetra is that layer.

One-line pitch we repeat on every slide: *"We don't measure pollution. We assign blame and trigger action."*

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4. Goals, Objectives & Success Metrics

4.1 Primary goal

Win **1st place** at ET AI Hackathon 2026 on PS5 by maximising the weighted judging score and beating every metric named in the official **Evaluation Focus**.

4.2 Product objectives

1. Attribute urban pollution to source categories at **ward / H3-cell** resolution with calibrated confidence.
2. Forecast AQI **24–72h ahead at ~1 km resolution** and **beat the persistence baseline** measurably.
3. Generate **prioritised, evidence-backed enforcement** recommendations an officer can act on.
4. Deliver **ward-level, multi-language citizen health advisories** across app / IVR / WhatsApp.
5. Prove the engine is **city-agnostic** by running it live on **≥3 cities**.

4.3 Success metrics (North Star + KPIs)

Metric	Definition	Target
North Star: Signal-to-Action latency	Time from an AQI/satellite signal to a validated, source-attributed enforcement recommendation	**< 5 minutes** (vs weeks/never today)
Forecast skill	RMSE improvement vs **persistence** baseline @24h	**≥ 25%** (stretch ≥ 35%); if realistic ~15–20%, report it honestly **and also beat climatology** — rigour beats inflated claims
Attribution agreement	Agreement of source split vs published inventory (SAFAR/TERI)	**Within ±15–20% per major category** (report honestly; even ±20% with confidence scores is credible)
Enforcement precision	**Rubric-proxy** (CPCB/GRAP-derived) "would-act" score on top-10 recommendations	**≥ 80%**
Language coverage	Citizen advisory languages supported	**≥ 4** (Hindi, English, Kannada, Marathi; Tamil = extensible stretch)
Scalability proof	Cities onboarded with **zero code change** (config only)	**≥ 3** live

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5. Win Strategy — Judging & Evaluation Alignment

The judging criteria are **identical across all 8 problem statements**. We engineer the product so each feature *visibly* earns points on a specific axis.

> 🏆 **Why VayuNetra wins #1 — in one breath**
>
> It is the **only** entry that turns India's air *data* into *action*: a satellite-ground **blame map** (Innovation) → a forecast that **provably beats persistence** with held-out backtests (Technical) → a **<5-minute signal-to-action** loop with cited, court-defensible enforcement (Business Impact) → a **city-agnostic engine live across 3 cities + a 4th onboarded on stage** (Scalability) → a **map-first console + multilingual citizen advisory** (UX). Built **100% free (₹0)** and **demo-proofed to never break live**. *Every metric the brief names, we put a number on a slide.*

5.1 Judging criteria → feature mapping

Criterion	Weight	How VayuNetra wins it
Innovation	**25%**	Satellite-ground **source "blame map"** with confidence scores; physics-informed ML; an *action engine*, not a dashboard. The framing — "assign blame, trigger action" — is itself the differentiator.
Business Impact	**25%**	India-scale: 1.67M deaths, 131 NCAP cities, ₹-quantified health/enforcement savings; signal-to-action latency cut from weeks → minutes; baseline-beating forecast quantified.
Technical Excellence	**20%**	Genuine **multi-agent orchestration** + **geospatial** engine + **RAG** + **atmospheric dispersion** + a forecast that **provably beats persistence** with held-out backtests.
Scalability	**15%**	**City-agnostic** data abstraction + H3 grid; **live multi-city** demo; cloud-native, config-driven onboarding; clear path to all 131 NCAP cities.
User Experience	**15%**	**Map-first** authority console (blame map, forecast time-slider, enforcement worklist) + **multi-language** citizen channel (app/IVR/WhatsApp). Beautiful, fast, role-aware.

5.2 Evaluation Focus → proof we will show (this is what separates #1 from the pack)

The brief's Evaluation Focus names five things. **Most teams will demo a UI. We will show evidence against each.**

Evaluation Focus item	Our concrete proof artifact
Source attribution accuracy vs **ground-truth emission inventories**	Side-by-side: our ward apportionment vs SAFAR/TERI study; agreement table + map overlay
AQI forecast accuracy at hyperlocal resolution (**RMSE vs persistence baseline**)	Backtest report: RMSE/MAE @24/48/72h, **our model vs persistence vs climatology**, with skill-score %
Enforcement recommendation quality **rated by domain experts**	Top-10 recommendations + auto-generated cited evidence dossiers, scored on a transparent **CPCB/GRAP-derived rubric proxy** (no expert needed; independent expert validation = planned next step — see ARCHITECTURE.md §14)
Citizen advisory **relevance & language coverage**	Live multi-language advisory (**Hindi / English / Kannada / Marathi**) + readability + native-speaker review
Demonstrated reduction in response time from signal to intervention	Timed demo: signal → attribution → enforcement packet in minutes; contrast with CAG-documented status quo

> **Strategic principle:** *If the brief names a metric, we will put a number on it on a slide.* That single discipline is our biggest edge.

6. Target Users & Personas

Persona	Role	Core "job to be done"	What VayuNetra gives them
Priya — City Pollution Control Officer (SPCB)	Decides daily air-quality response	"Where is the problem coming from and where do I send my 4 inspectors today?"	Blame map + ranked enforcement worklist + evidence dossiers
Rajan — Municipal Commissioner / Smart City CEO	Policy & accountability	"Are our interventions working? How do we compare to other cities?"	Multi-city comparative dashboard + intervention effectiveness tracking
Field Inspector	Executes enforcement	"Is this site actually the culprit, and what's my legal basis?"	Geo-tagged evidence packet + regulatory citation (RAG)
Citizen (incl. vulnerable groups)	Self-protection	"Is the air dangerous for my child / my outdoor job today, in my language?"	Ward-level, multi-language advisory via app / IVR / WhatsApp
CPCB / Policy analyst	National oversight	"Which cities, which sources, which seasons need funding & rules?"	Cross-city trends, compliance metrics, source patterns

Primary buyer/user for the demo: the **City Pollution Control Officer** + **Municipal Commissioner** — the people who can *act*. Citizens are the impact multiplier.

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7. Product Scope

7.1 MVP (must exist for a credible win)

- City-agnostic data ingestion for **≥3 cities** (CAAQMS + satellite + weather + land use).
- **Source Attribution Agent** (Agent 1) producing ward/H3 apportionment + confidence.
- **Hyperlocal Forecast Agent** (Agent 2) @24–72h, ~1 km, **with persistence baseline comparison**.
- **Authority console**: blame map + AQI layer + forecast time-slider.
- **Enforcement Agent** (Agent 3): ranked worklist + auto evidence dossier (RAG-cited).
- **Citizen Advisory Agent** (Agent 4): ward alerts in **≥3 languages**, at least app + IVR/WhatsApp text.
- Backtest/validation report (the evidence artifacts in §5.2).

7.2 Finale upgrades (to lock #1)

- **Multi-City Comparative Intelligence** (Agent 5) + intervention-effectiveness analytics.
- Deep-learning forecast upgrade (spatiotemporal GNN / TFT) over the gradient-boosting MVP.
- Full atmospheric **dispersion modelling** (Gaussian plume + wind-field advection) as physics prior.
- **Enhanced AI models (§8 ⚡):** Satellite Vision source detection (E1), Dense-coverage AOD→PM2.5 + 1km downscaling (E2), What-if intervention simulator (E3); **prescriptive optimiser (E5), multimodal satellite-evidence dossiers (E6), health & carbon quantification (E7)**; spike/anomaly detector (E4, stretch).
- Live cloud deployment with a public demo URL + IVR phone number that judges can call.
- Polished demo video + pitch deck + architecture diagram.

7.3 Out of scope (explicitly)

- Deploying new physical sensors / hardware (we consume existing CAAQMS + satellite).
- Regulatory/legal e-filing integration with government systems (we *generate* the dossier; filing is future).
- Nationwide rollout to all 131 cities during the hackathon (we *prove* the path with 3).

7.4 Future roadmap (post-hackathon, shows vision)

- Indoor air + wearable integration; hospital-admission correlation; carbon co-benefit tracking; private-sensor crowd network fusion; full GRAP automation; SPCB system integrations.

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8. Core Features — The Five Agents

> All five map directly to the PS5 "What You May Build" list. Each spec includes inputs, outputs, method, the **demo moment**, and **acceptance criteria**.

Agent 0 — Orchestrator / Supervisor (the multi-agent backbone)

- **Purpose:** Coordinate the specialist agents, maintain shared state (city, time window, ward/H3 cell), enforce human-in-the-loop gates, and expose a single conversational + API surface.
- **Method:** Stateful agent graph (LangGraph) with typed shared memory; tool-calling; guardrails; trace logging for auditability.
- **Why it matters:** This is what makes us a **genuine multi-agent system** (what the organizers signal across every PS), not a pipeline of scripts.

Agent 1 — Geospatial Pollution Source Attribution Engine ⭐ (our hero)

- **What it does:** Attributes pollution to **source categories** — `{vehicular/traffic, construction & road dust, industrial, biomass/waste burning, regional/transported, other}` — per **ward / H3 cell**, with **statistical confidence scores**.
- **Inputs:** CAAQMS pollutant mix (PM2.5, PM10, NO_2, SO_2, CO, O_3), **Sentinel-5P** (NO_2, SO_2, CO, aerosol index), **MODIS/VIIRS** AOD + active-fire/thermal anomalies, **Sentinel-2** land-use & construction-site change, traffic density, OSM land use, industrial/permit registries, wind field.
- **Method: Physics-informed ML** — chemical-signature priors (e.g., NO_2↑ → combustion/traffic; SO_2↑ → industrial; AOD + fire pixels → biomass burning; coarse-PM/PM10:PM2.5 ratio → dust/construction) + a learned apportionment model calibrated against published source-apportionment studies; spatial smoothing over the H3 grid.
- **Output:** Per-cell source-share vector + confidence; the **"blame map"**.
- **Demo moment:** Click a red ward → *"68% construction dust, 22% traffic, 10% transported — confidence 0.83,"* with the satellite + sensor evidence that proves it.
- **Acceptance:** Apportionment within **±15%** per major category vs SAFAR/TERI on validation wards.

Agent 2 — Hyperlocal Predictive AQI Forecasting Agent ⭐ (our proof-of-rigor)

- **What it does:** Forecasts AQI **24–72h ahead at ~1 km (H3 res 8 ≈ 0.7 km²) grid** across the whole city.
- **Inputs:** Meteorological forecasts (wind speed/direction, boundary-layer height, temp, humidity, precip — IMD/GFS/ERA5/OpenWeather), traffic & **seasonal emission calendars** (Diwali, stubble-burning windows, winter inversion), persistence & climatology features, **atmospheric dispersion** output as a physics feature.
- **Method (staged):** MVP = gradient-boosting (LightGBM/XGBoost) spatiotemporal model with engineered met + lag features; **Finale = spatiotemporal GNN / Temporal Fusion Transformer**. **Baselines = persistence + climatology.** Prediction intervals via quantile regression.
- **Output:** Gridded 24/48/72h AQI forecast + uncertainty bands + "spike alerts."
- **Demo moment:** Time-slider scrubs 72h forward; a spike lights up tomorrow 6 PM in a ward → triggers a pre-emptive advisory + enforcement pre-position.

- **Acceptance: ≥25% RMSE reduction vs persistence @24h** on a held-out test period.

Agent 3 — Enforcement Intelligence & Prioritisation Agent

- **What it does:** Correlates current/forecast hotspots + attribution with **registered emission sources** (industries, construction sites, waste-burning spots, diesel corridors) → **prioritised, evidence-backed enforcement worklist**.
- **Inputs:** Attribution map (Agent 1), forecast (Agent 2), source registries (SPCB consent-to-operate, construction permits, OSM industrial zones, fire pixels), population-exposure layer, **RAG** over CPCB/SPCB regulations + NCAP + GRAP.
- **Method:** Exposure-weighted prioritisation score = `f(source contribution × population exposed × actionability × confidence)`; LLM-generated **evidence dossier** with regulatory citations (RAG, with sources).
- **Output:** Ranked inspector deployment list + per-target **evidence packet** (what, why, satellite/sensor proof, legal basis).
- **Demo moment:** *"Send inspectors to these 3 construction sites first — together they drive 41% of ward-12 PM2.5; here's the auto-generated notice with the CPCB rule cited."*
- **Acceptance: ≥80%** "would-act" on top-10 via a transparent **CPCB/GRAP rubric proxy** (ARCHITECTURE.md §14); independent expert validation = planned next step.

Agent 4 — Citizen Health Risk Advisory Agent

- **What it does:** Generates **ward-level health-risk alerts**, maps **population vulnerability** (hospitals, schools, outdoor workers, elderly) against **forecast** AQI, and pushes **personalised advisories** via mobile app, public displays, and **IVR in regional languages**.
- **Inputs:** Forecast (Agent 2), vulnerability layers (Census/WorldPop, OSM hospitals/schools), AQI→health breakpoints (CPCB/WHO), user profile (location, sensitivity).
- **Method:** Rule-based health-risk tiering + **LLM localisation/translation** for natural, culturally-appropriate messaging; channel adapters (PWA push, SMS, WhatsApp, IVR text-to-speech).
- **Output:** *"Bengaluru, Ward 84, tomorrow AM: AQI 'Very Poor'. Outdoor workers — shift heavy work to after 4 PM"* — delivered in **Kannada** (Bengaluru), **Marathi** (Mumbai), **Hindi/English** (Delhi); architecture extensible to all 22 scheduled languages.
- **Demo moment:** Judge calls a **live IVR number** / messages the **Telegram bot**, hears or reads a real Kannada advisory.
- **Acceptance:** ≥4 languages (hi/en/kn/mr); native-speaker-rated relevance; correct vulnerability targeting.

Agent 5 — Multi-City Comparative Intelligence

- **What it does:** Tracks & compares AQI trends, **intervention effectiveness**, and compliance across cities; recommends "what worked in city A for city B."
- **Inputs:** All cities' historical + live data, intervention logs.
- **Method:** Causal-ish before/after intervention analysis + benchmarking; comparative geo-analytics.
- **Output:** Cross-city dashboard + intervention playbook recommendations.
- **Demo moment:** *"Delhi's construction-dust crackdown cut ward PM2.5 by X%; Bengaluru's ward-84 has the same signature — apply the same playbook."*
- **Acceptance:** ≥3 cities compared; at least one quantified intervention-effectiveness story.

⚡ Enhanced AI Models (finale upgrades — the team's CV/ML firepower)

> These raise **Technical Excellence + Innovation** and fill real data gaps. Scope = **finale upgrades** layered on the core 5 agents (balanced scope); E4 is the lowest-priority stretch. Full training detail in §12.

E1 — Satellite Vision: Source Detection (trained CV) ⭐

- **What:** A trained CNN / semantic-segmentation model on free **Sentinel-2 / Landsat** imagery that detects **active construction sites, brick kilns, and open-burning scars**.
- **Why it matters:** Open construction-permit data barely exists per city — this model *generates* that missing layer from satellite, feeding Agent 1 (attribution) and Agent 3 (enforcement). Closes our biggest data gap and showcases the team's CV strength.
- **Demo moment:** Toggle the "detected sources" layer → freshly-detected construction sites light up and auto-populate the enforcement worklist.

E2 — Dense-Coverage ML: AOD→PM2.5 + 1km Downscaling (trained) ⭐

- **What:** (a) a model mapping satellite **Aerosol Optical Depth → ground PM2.5**, and (b) an ML **downscaling/super-resolution** model fusing sparse stations + satellite into a dense **1km** field.
- **Why it matters:** Delivers AQI **everywhere**, not just at the ~40 stations a city has — this is what makes "hyperlocal at 1km" *real* rather than interpolated guesswork.
- **Demo moment:** Switch from "stations only" (sparse dots) to "VayuNetra dense grid" (full-city 1km heatmap).

E3 — Intervention What-If Simulator ⭐

- **What:** Reuses the forecast + dispersion engines to simulate interventions — *"halt construction in ward X / divert this diesel corridor → projected AQI change tomorrow."*
- **Why it matters:** Turns the platform from *predictive* into *prescriptive* decision-support — exactly the "reduce pollution **at source**" intent of the brief. A killer interactive demo moment.
- **Human-impact quantification (Business-Impact gold):** every simulated intervention reports **"≈ N people protected · ≈ T tonnes PM2.5 avoided · exposure-hours reduced"** by overlaying the ΔAQI on the WorldPop population layer — turning an abstract AQI drop into lives and tonnes. *(E7 extends this with ₹ health-cost savings, respiratory cases prevented, and CO_2e co-benefit.)*
- **Demo moment:** Officer toggles *"pause construction, Ward 12"* → the map re-forecasts, AQI drops 'Very Poor' → 'Moderate', and a card reads *"≈ 18,000 people protected · ≈ 2.3 t PM2.5 avoided tomorrow."*

E4 — Spike / Anomaly Detector (trained, stretch)

- **What:** A trained model that flags abnormal pollution events (a new illegal burn, an industrial upset) against the normal diurnal/seasonal baseline.
- **Why it matters:** Catches the *novel* event a static threshold misses — feeds proactive enforcement. Lowest-priority stretch.

E5 — Prescriptive Intervention Optimiser ⭐ (extends E3)

- **What:** Beyond single what-if toggles, a constrained optimiser recommends the **best bundle of 2–3 interventions** (e.g., *halt construction in Ward 12 + divert diesel corridor X + inspect 4 CV-detected sites*) that maximises AQI reduction **under an inspector-hour / budget limit**.
- **Why it matters:** It answers the officer's *real* question — *"what's the best use of my 4 inspectors tomorrow?"* — moving the platform from predict → prescribe → **optimise**. Almost every other team stops at single-toggle what-if. **Innovation + Business Impact.**
- **Method:** greedy / priority-knapsack search over the E3 simulator (dispersion + forecast), scoring each candidate by exposure-weighted ΔAQI per unit resource; returns **top-3 ranked action packages** with trade-

offs. (No new model — search over the existing engines.)

- **Demo moment:** *"Optimise for tomorrow's spike under 20 inspector-hours"* → map shows the recommended package: *"−18 AQI · protects 42k people · 12 inspector-hours."*

E6 — Multimodal RAG: Satellite Visual Evidence ⭐

- **What:** Enforcement dossiers cite not just regulations but the **actual Sentinel-2 image patch** showing the offending construction/burning, via CLIP-style image embeddings stored alongside the text RAG.
- **Why it matters:** Multimodal RAG is rare in hackathons; it makes enforcement **visceral and court-defensible** — the officer (and the judge) *sees* the satellite proof next to the cited rule. **Innovation + Technical Excellence.**
- **Method:** a free CLIP / vision encoder (sentence-transformers) embeds Sentinel-2 patches → pgvector; the dossier retrieves the most relevant patch + the governing CPCB/GRAP rule. Pretrained encoder — no custom training.
- **Demo moment:** open a dossier → *"Here is the Sentinel-2 patch from 2 days ago showing the active dust plume,"* beside the cited regulation.

E7 — Health & Carbon Impact Quantification ⭐

- **What:** A quantification layer that converts every attribution / forecast / what-if / optimiser output into **public-health and carbon terms**: ≈ *₹X crore health costs avoided · N asthma & respiratory exacerbations prevented · T tonnes CO_2e co-benefit*, using standard dose-response (WHO/CPCB) and emission factors.
- **Why it matters:** ET / finance judges reward **rupee impact**; it bridges the **Sustainability** theme (national net-zero) on top of Smart Cities. Near-zero build cost (static, *cited* factor tables). **Business Impact.**
- **Demo moment:** every what-if / optimiser card and citizen advisory shows ₹ + lives + CO_2e, not just an AQI delta.

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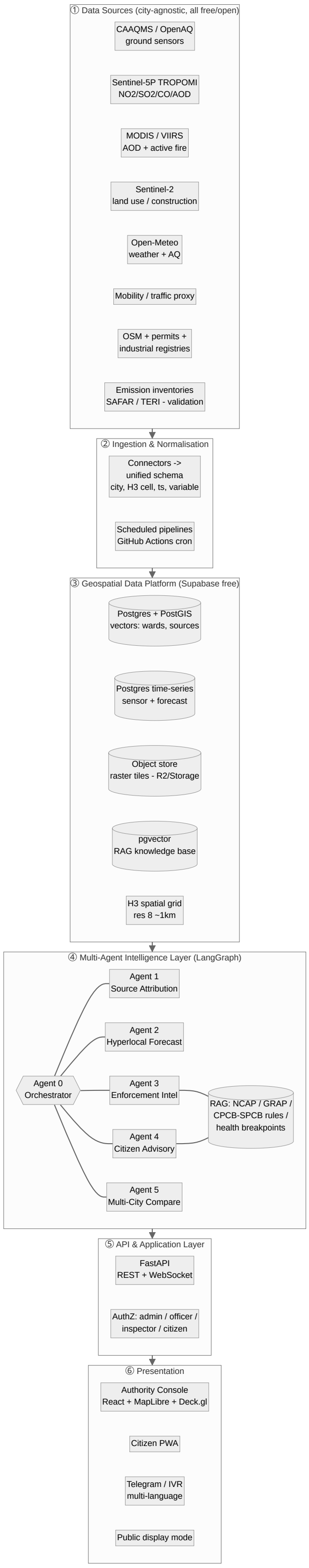
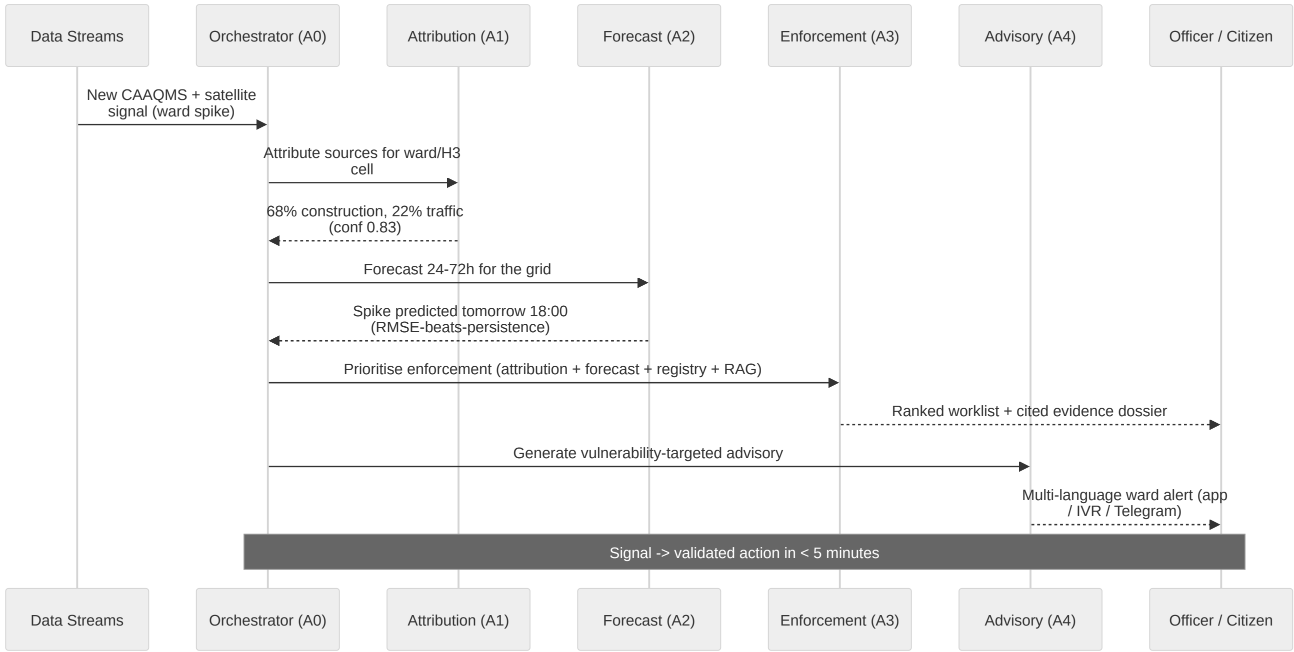
9. System Architecture

9.1 High-level architecture (city-agnostic, multi-agent)



9.2 Agent orchestration flow (signal → action)



9.3 Architectural principles

- **City-agnostic by construction:** every connector normalises to `(city_id, h3_cell, timestamp, variable, value, source, confidence)`. Adding a city = a config + data-source registration, **no code change**.
- **Geospatial-native:** H3 hexagonal grid (res 8 ≈ ~1 km) as the universal spatial key → clean joins across sensors, satellite, population, sources.
- **Physics + ML hybrid:** dispersion modelling and chemical priors keep the ML honest and explainable (critical for enforcement/legal credibility).
- **Auditable agents:** every agent action is traced and citable (RAG sources) — essential for enforcement defensibility and judge trust.
- **Human-in-the-loop:** enforcement actions are *recommended*, never auto-executed; officer approves.

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10. Technology Stack

Layer	Choice	Why
Language	Python (core), TypeScript (frontend)	Geospatial + ML ecosystem; team is full-stack
Agent framework	**LangGraph** (stateful multi-agent)	Controllable, traceable, typed state; better than ad-hoc chains
LLM	**Free-tier: Google Gemini Flash** (multilingual, RAG, localisation); fallbacks Groq / Ollama. Swap to Claude if the hackathon grants credits	Model-agnostic via LangGraph; $0 default
Satellite / RS	**Google Earth Engine** (Sentinel-5P/2, MODIS/VIIRS)	Free, planetary-scale, no raw-tile wrangling
Geospatial	PostGIS, **Uber H3**, GeoPandas, rasterio, Deck.gl	Industry-standard spatial stack
Database	**Supabase** Postgres 15 + **PostGIS** + **pgvector** (native partitioning for time-series)	Free tier; one DB for relational + spatial + vectors + time-series

Layer	Choice	Why
ML / forecasting	LightGBM/XGBoost (MVP) → PyTorch (GNN/TFT, finale); scikit-learn	Fast baseline → deep upgrade path
Dispersion	Gaussian plume (custom/AERMOD-style) + HYSPLIT/wind-advection	Physics prior + explainability
Backend/API	FastAPI + WebSocket	Async, fast, typed
Pipelines / scheduler	**GitHub Actions** cron	Free scheduled data refresh + forecast / agent runs
Frontend	React + **MapLibre GL + Deck.gl** + Tailwind (Vercel)	Free, token-free maps; best-in-class geo viz
Citizen channels	**Free:** Telegram bot + PWA; IVR demo via Twilio trial; WhatsApp = production upgrade	Reach beyond smartphones (IVR for low-tech users)
MLOps/Infra	**Free-first:** Supabase (Postgres/PostGIS/pgvector/Auth) · Cloud Run free · Vercel · GitHub Actions cron · Colab/Kaggle (train) · Earth Engine (free)	$0 infra; see ARCHITECTURE.md §5
Observability	Grafana + structured agent traces	Demo the "signal→action" timing live

> **Infra = 100% free-tier (₹0).** Google Earth Engine is free for non-commercial / hackathon use; hosting via Supabase + Cloud Run free + Vercel + GitHub Actions. This is also a pitch asset — any of 131 NCAP cities can adopt VayuNetra at near-zero infra cost. Full breakdown in ARCHITECTURE.md §5. Architecture stays cloud- and model-agnostic.

11. Data Sources & Data Strategy

> **Everything below is publicly accessible** — matching the team's "mostly public data" reality. This is a major de-risking advantage of PS5.

Data	Source / Access	Use	Cadence
Ground AQI (PM2.5/10, NO_2, SO_2, CO, O_3)	**CPCB CAAQMS** (data.gov.in / CPCB dashboard), **OpenAQ** API (fallback)	Truth signal, attribution, forecast target	~Hourly
NO_2/SO_2/CO/Aerosol columns	**Sentinel-5P TROPOMI** via Earth Engine	Attribution (combustion/industrial signatures)	Daily passes
Aerosol Optical Depth + active fire	**MODIS / VIIRS** via Earth Engine	Biomass/waste-burning attribution; PM proxy	Daily
Land use / construction change	**Sentinel-2** via Earth Engine, OSM	Construction-dust & industrial attribution	5-day / static
Meteorology + forecasts	**Open-Meteo** (free, no key — incl. Air-Quality API), IMD, ERA5/GFS	Forecast features, dispersion	Hourly/forecast
Mobility feeds (PS5-named)	**Primary (free):** OSM road network + GTFS transit + time-of-day/day-of-week traffic model + Mapbox/TomTom free-tier where available	Vehicular & diesel-corridor attribution + forecast feature	Real-time / engineered proxy
Industrial & permit registries	SPCB consent-to-operate lists, city construction permits, OSM industrial polygons	Enforcement targeting	Static/periodic

Data	Source / Access	Use	Cadence
Emission inventories (**validation**)	**SAFAR**, **TERI**, REAS, EDGAR, city apportionment studies	Ground-truth for attribution accuracy	Reference
Population & vulnerability	Census, **WorldPop**, OSM (hospitals/schools)	Exposure weighting, advisory targeting	Static
Regulations & SOPs (**RAG**)	NCAP, **GRAP**, CPCB/SPCB rules, WHO/CPCB AQI health breakpoints	Enforcement basis + advisory thresholds	Static corpus

Data strategy notes

- **Cold-start without live gov APIs:** if a city's live feed is flaky, we backfill from OpenAQ + historical CPCB archives so the demo never breaks.
- **Confidence everywhere:** every fused value carries a provenance + confidence so attribution/forecast outputs are defensible.
- **Validation data is sacred:** SAFAR/TERI inventories are held out *only* for evaluation, never used in training — so our accuracy claims are honest.

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12. AI / ML Methodology

12.0 Models & Training Plan (overview)

> **Yes — this project trains real models.** The table is every trainable component, its type, training data, and free compute. The **forecast model is the hero** — it is *how we beat the persistence baseline*, the single most important number on the scoreboard. Training runs free on **Google Colab / Kaggle**; artifacts are versioned to Storage/R2; a single reproducible `evaluate.ipynb` regenerates every metric for the judges.

Model	Type	Trained on	Free compute	Serves
Forecast ⭐	LightGBM (MVP) → spatiotemporal GNN/TFT	historical CAAQMS + met + seasonal calendars	Colab/Kaggle GPU	Agent 2 — beats persistence
Source attribution	Gradient boosting (multi-output) + SHAP	multi-pollutant + satellite + land use, calibrated to SAFAR/TERI	Colab	Agent 1 — blame map
Satellite source CV (E1)	CNN / segmentation (transfer-learned)	labelled Sentinel-2 tiles (construction/kiln/burn)	Kaggle GPU	Agents 1 & 3
AOD→PM2.5 (E2)	Regression (GBM / small NN)	paired satellite AOD + ground PM2.5	Colab	dense coverage
1km downscaling (E2)	CNN / learned interpolation	sparse stations + satellite + land use	Kaggle GPU	hyperlocal 1km
Spike / anomaly (E4)	STL + isolation-forest / autoencoder	historical per-cell series	Colab	proactive alerts

(The RAG retrieval and citizen-advisory LLM use ***pretrained*** *models — no custom training needed there; that is the correct, efficient choice, not a gap.)*

12.1 Source attribution (Agent 1)

- **Hybrid, explainable approach:**

1. **Chemical-signature priors** — ratios & tracers: `PM10:PM2.5` (dust vs combustion), NO_2 (traffic/combustion), SO_2 (industrial/coal), CO (incomplete combustion), satellite AOD + fire pixels (biomass), O_3 (secondary/photochemical).
2. **ML apportionment** — gradient-boosting / spatiotemporal model mapping the multi-pollutant + satellite + land-use feature vector → source-share vector per H3 cell.
3. **Calibration & confidence** — calibrate against published apportionment; output per-category confidence; spatial smoothing across neighbouring cells.

- **Explainability:** SHAP-style feature attribution so an officer sees *why* a ward was blamed on construction vs traffic.

12.2 Hyperlocal forecasting (Agent 2)

- **Targets:** AQI / PM2.5 per H3 cell at +24/+48/+72h.
- **Features:** met forecasts (wind, BLH, precip, temp, RH), lagged AQI, **dispersion-model output**, traffic, seasonal/event calendars (stubble windows, Diwali, winter inversions), land use, day-of-week/hour.
- **Models: Baseline = persistence & climatology** (mandatory, because the brief grades against persistence). **MVP model = LightGBM** with quantile loss for intervals. **Finale = spatiotemporal GNN / Temporal Fusion Transformer** sharing information across stations + grid.
- **Uncertainty:** quantile regression → prediction intervals + spike-probability.

12.3 Atmospheric dispersion (physics prior) — *this makes our forecast physics-informed*

- **Gaussian plume** for local point/area sources + **wind-field advection** of satellite NO_2/AOD for regional transport; optional **HYSPLIT** back-trajectories to attribute transported pollution. Feeds both attribution (transported share) and forecasting (physics feature).
- **Innovation framing (use it on the deck):** because the dispersion physics is injected as a feature/constraint into the ML forecast, this is a genuine **physics-informed ML** hybrid — not a black-box regressor. It's explainable *and* grounded in atmospheric science, which is exactly the "novel ML" depth judges look for, **without** the execution risk of a full from-scratch PINN.
- **Optional stretch (not committed):** a true **Physics-Informed Neural Network (PINN)** coupling the advection-diffusion equation into the loss. High Technical-Excellence upside but high risk for a 2–3 person team — pursue *only* if the core is flawless and time remains.

12.4 Enforcement prioritisation (Agent 3)

- **Score:** `priority = source_contribution × population_exposed × actionability × confidence`, computed per candidate source; ranked.
- **RAG dossier:** retrieve the governing rule (CPCB/SPCB/NCAP/GRAP), synthesise a cited, court-defensible evidence packet.

12.5 Citizen advisory (Agent 4)

- **Health tiering** from CPCB/WHO AQI breakpoints × vulnerability class; **LLM localisation** into regional languages with culturally-appropriate, actionable phrasing; channel adapters (push/SMS/WhatsApp/IVR-TTS).

12.6 Multi-city comparison (Agent 5)

- **Before/after intervention deltas** with seasonality control; cross-city signature matching to recommend proven playbooks.

12.7 Satellite source-detection CV (E1)

- **Approach:** label a set of **Sentinel-2** tiles for construction sites / brick kilns / open-burning, transfer-learn from a pretrained backbone (U-Net / ResNet encoder, optionally Sentinel-1 SAR for cloud-robustness); output geo-located detections → `emission_sources`.
- **Why trainable, not rule-based:** spectral + textural signatures of construction/burning are exactly what CNNs excel at; fixed thresholds fail across seasons and cities.

12.8 Dense-coverage models (E2)

- **AOD→PM2.5:** GBM/MLP mapping satellite AOD + met → surface PM2.5 (the established remote-sensing approach), calibrated on station pairs.
- **1km downscaling:** learned spatial super-resolution fusing sparse stations + satellite + land use → dense H3-res-8 field with uncertainty. Turns ~40 stations into a full-city 1km map.

12.9 Intervention what-if simulator (E3)

- **Method:** counterfactual run of the dispersion + forecast models with a source switched off/reduced → projected ΔAQI per ward + confidence. Prescriptive, not just predictive.
- **Impact quantification:** ΔAQI × population (WorldPop) × exposure-response → **people protected + PM2.5 tonnes avoided + exposure-hours reduced** per scenario (the numbers that win Business Impact).

12.10 Spike / anomaly detector (E4, stretch)

- **Method:** model the normal diurnal/seasonal baseline (STL + isolation forest / autoencoder); flag deviations as candidate events for proactive enforcement.

12.11 Responsible AI & fairness

- **Equity guard:** enforcement prioritisation must not systematically over-target low-income wards — we monitor the exposure-weighting for fairness and keep a **human-in-the-loop** approval gate.
- **Honesty:** every output carries a confidence score; validation data (SAFAR/TERI) is **never** trained on; we report real skill scores, not cherry-picked wins.
- **Privacy:** citizen advisory uses **ward-level**, not individual, location by default.

12.12 Training compute & MLOps (free)

- **Train:** Google Colab (free T4) + Kaggle (30h/wk GPU); checkpoint to Storage/R2. **Track:** lightweight MLflow / logged metrics; versioned artifacts. **Reproducibility:** one `evaluate.ipynb` regenerates every metric (no leakage, fixed seeds, inventories held out).

12.13 Prescriptive optimiser (E5)

- **Problem:** given forecast + attribution + (E1) detected sources, choose the subset of interventions that maximises exposure-weighted ΔAQI subject to an inspector-hour / budget constraint.
- **Method:** greedy / priority-knapsack over candidate interventions, each scored by simulated ΔAQI × population exposed ÷ resource cost (via the E3 engine); return top-3 ranked packages with trade-offs. **No model training** — search over the existing physics + ML simulators.

12.14 Multimodal RAG — satellite visual evidence (E6)

- **Method:** a free **CLIP** / vision encoder embeds Sentinel-2 patches around each active/detected source into the same pgvector store (`modality='image'`). The enforcement dossier retrieves the most relevant patch **+** the governing CPCB/GRAP rule, citing both. Pretrained encoder — no custom training.

12.15 Health & carbon quantification (E7)

- **Method:** static, **citable** factor tables — WHO/CPCB **dose-response** (ΔPM2.5 → attributable cases / mortality risk → ₹ health cost via standard valuation) and **emission factors** (source reduction → CO_2e co-benefit) — applied to attribution, forecast, and what-if/optimiser outputs. **Every figure cites its factor source** (honest, never an invented constant).

13. Validation & Evaluation Plan

> This section is our **#1 differentiator**. We treat the brief's Evaluation Focus as a test suite.

#	What we validate	Method	Reported as
1	**Attribution accuracy**	Compare ward apportionment to SAFAR/TERI held-out inventory	Per-category agreement table + map overlay; "within ±X%"
2	**Forecast skill**	Time-split backtest (train past → test held-out weeks); compute RMSE/MAE @24/48/72h for **model vs persistence vs climatology**	Skill score = `1 − RMSE_model/RMSE_persistence`; target ≥0.25
3	**Enforcement quality**	Score top-10 on a transparent **CPCB/GRAP-derived rubric proxy** (correct source, actionable, defensible, cited) — no domain expert required	% "would-act" ≥80%
4	**Advisory relevance & coverage**	Native-speaker review; count languages; readability	≥4 languages (hi/en/kn/mr); relevance score
5	**Signal-to-action latency**	Instrument the pipeline; time signal → enforcement packet	Median < 5 min; contrast with CAG status quo
6	**Satellite source-detection CV** (E1)	Precision/recall on held-out labelled tiles	mAP / F1 on construction & burning detection
7	**Dense-coverage models** (E2)	AOD→PM2.5 RMSE; downscaling skill vs plain interpolation	RMSE / skill score at held-out stations
8	**What-if simulator plausibility** (E3)	Sanity vs dispersion physics + historical analogues	directional correctness + magnitude sanity
9	**Fairness / equity audit**	Partial correlation of enforcement priority with ward income, *controlling for* actual source contribution + exposure	≈ 0 — income adds **no independent** targeting signal (priority is driven by pollution, not poverty)
10	**Optimiser quality** (E5)	Compare the optimiser's package vs best-single-action + random baselines on simulated ΔAQI per inspector-hour	Δ improvement over single-action baseline
11	**Multimodal retrieval** (E6)	CLIP patch-retrieval relevance on a held-out labelled tile set	precision@k of the cited image evidence
12	**Impact factors** (E7)	Every ₹ / health / CO_2e figure traceable to a cited WHO/CPCB/emission factor	100% sourced — no invented constants

Backtesting discipline: strict temporal splits (no leakage), validation inventories never trained on, fixed random seeds, and a reproducible evaluation notebook we can show judges live.

14. UX / Product Design

14.1 Authority Console (primary, web)

- **Map-first** (Deck.gl). Default view = the **Blame Map**: H3/ward choropleth coloured by *dominant source*, not just AQI.
- **Layers toggle:** live AQI · source attribution · 24-72h forecast (time-slider) · enforcement targets · population vulnerability · satellite overlays.
- **Right rail = Action panel:** ranked enforcement worklist; click a target → evidence dossier with citations + **"Generate Cited Dossier (PDF export)"** + "Dispatch / Generate Notice."
- **Top bar:** city switcher (proves multi-city), time scrubber, scenario toggle ("now" vs "tomorrow 18:00"), and a **live "Signal → Action: 2m 47s" latency widget** (visceral North-Star proof on screen).
- **Key toggles (the demo wow): "Detected Sources"** (E1 satellite-CV construction/kiln/burn) · **"Stations-only ↔ Dense 1km grid"** (E2) · **"What-if intervention"** (E3) · **SHAP tooltips** on the blame map (*"NO_2 + AOD drove 68% construction blame"*).
- **Comparative tab:** cross-city benchmarking + intervention effectiveness.
- **Fairness panel:** enforcement-action distribution across socio-economic wards — pre-empts any "are you over-targeting the poor?" question with a transparent answer.

14.2 Citizen channel

- **PWA:** ward AQI now + 72h, personalised health action, vulnerability-aware ("you flagged an asthmatic child").
- **Telegram bot + IVR:** multi-language (Hindi/English/Kannada/Marathi). **A judge can call a live number or message the bot.**
- **Public display mode:** big-screen ward board for bus stops/schools.

14.3 Design principles

- Every screen answers **"what do I *do*?"**, not just "what is the number."
- Confidence shown, never hidden. Sources cited. Fast (<2s map interactions). Mobile-first citizen, desktop-first authority.

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15. Scalability & Multi-City Design

Scalability is 15% of the score and a core team-chosen requirement ("multi-city from day one"). We make it visible.

- **Config-driven onboarding:** a new city = register its CAAQMS endpoints + boundary polygon; H3 grid + satellite + weather are global by default → **zero code change**.
- **Universal spatial key (H3):** identical pipeline math for every city.
- **Stateless, containerised agents:** horizontal scale on Cloud Run/GKE; per-city pipelines run in parallel.
- **Demo proof:** run **Delhi + Bengaluru + Mumbai** live and switch between them on stage; show a 4th city onboarded *during* the demo from config alone.
- **Path to scale narrative:** 3 cities today → **131 NCAP non-attainment cities** with the same engine.

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16. Implementation Roadmap (Phase-Gated)

> **Time is not the constraint** — phases are gated by *capability*, not dates. A 2–3 person team executes them largely in sequence with the lead parallelising data + frontend early.

Phase	Goal	Key outputs	Exit gate
P0 — Foundation	Data spine + city-agnostic schema	Ingestion for 3 cities, H3 grid, PostGIS (Supabase), base map UI	Live AQI renders on the map for 3 cities
P1 — Attribution MVP	Agent 1 + blame map	Source apportionment + confidence + validation vs inventory	"Blame map" demo + ±15% agreement shown
P2 — Forecast	Agent 2 + baseline beat	24-72h gridded forecast + **persistence backtest**	≥25% RMSE skill over persistence
P3 — Action layer	Agents 3 & 4 + RAG	Enforcement worklist + dossiers; multi-language advisory + IVR	Signal→action <5 min; live IVR call works
P4 — Scale & compare	Agent 5 + 4th-city onboarding	Comparative dashboard; config-only new city	3 cities compared + live onboard demo
P4.5 — AI model upgrades	E1 Satellite CV + E2 dense-coverage + E3 what-if + **E5 optimiser + E6 multimodal evidence + E7 health/carbon**; E4 spike if time	Trained models live; what-if re-forecasts; optimiser returns ranked packages	CV detections feed enforcement; dense 1km grid renders; dossiers cite satellite patches; ₹/health/CO_2e on every card
P5 — Win polish	Deliverables	Architecture diagram, deck, demo video, deployed URL	Dry-run pitch scores 5/5 on internal rubric

Parallelisation for a small team: Person A = data/geo + dispersion; Person B = ML (attribution + forecast) + validation; Person C (or shared) = agents/LLM + frontend + citizen channels + pitch. Lead owns the demo narrative end-to-end.

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17. Team & Roles

Role	Owner	Responsibilities
Lead / Demo owner	TBD	Architecture, orchestrator, demo narrative, pitch, judge Q&A
Data & Geospatial Eng	TBD	Connectors, Earth Engine, PostGIS/H3, dispersion model, scalability
ML / Forecasting	TBD	Attribution model, forecast model, **baseline backtests**, validation report
Agents / Frontend / Citizen	shared	LangGraph agents, RAG, React/MapLibre/Deck.gl console, Telegram/IVR, multi-language

(2–3 people wear multiple hats; the lead must personally own the validation numbers and the demo story — those win the room.)

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18. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
"This is just another dashboard" perception	Med	**High**	Relentless *action-engine* framing; lead with attribution + enforcement, never with an AQI gauge
Live gov data feeds flaky during demo (the #1 risk)	Med	**High**	**"Demo Mode": a frozen, versioned, deterministic snapshot** of all 3 cities (ground + satellite features + precomputed attribution/forecast/enforcement/advisories) bundled in the repo; one `DEMO_MODE=true` flag runs the *entire scored demo offline* — zero live-API dependency. The live system is *also* shown, but the demo cannot break. Plus OpenAQ fallback + pre-warm.
Attribution hard to validate convincingly	Med	High	Calibrate to published SAFAR/TERI; show agreement honestly; confidence scores
Forecast doesn't beat persistence enough	Low–Med	High	Strong met features + dispersion prior; report honest skill score; even 20% wins if rigorous
Scope too big for 2–3 people / "jack of all trades, master of none"	**High**	Med	**Core-first discipline:** the 5 agents + baseline-beating forecast must be flawless *before* any enhancement. E1–E4 are **finale-only** upgrades, each independently shippable — if time runs short we cut E4→E3→E2 and still have a complete, winning core. Phase gates enforce this.
Multi-language quality (Kannada/Tamil/Marathi)	Med	Med	Native-speaker review; LLM + human-in-loop check; keep messages short/templated
Compute for deep models	Low	Med	Gradient-boosting MVP first; GNN/TFT only if time/compute allow
Enforcement legal defensibility doubted	Low	Med	RAG citations + human-in-loop approval + audit trail

19. Deliverables & Demo Narrative

19.1 Required deliverables (per PS5) — elevated to win

Deliverable	How we make it #1-grade
Working Prototype	Live, cloud-deployed URL + **callable IVR number**; 3 cities switchable; real data
Architecture Diagram	The mermaid diagrams in §9, rendered cleanly; shows multi-agent + geo + RAG + dispersion
Presentation Deck	Problem (1.67M deaths) → wedge → live blame map → baseline-beating numbers → multi-city → impact → ask
Demo Video	The narrative below, ≤3 min, tight, emotional open + metric-driven close

19.2 Demo / pitch narrative (the storyline that wins the room)

1. **Hook (emotional):** an outdoor worker / child in a red-AQI ward. *"1.67 million Indians die from this every year — and our cities still only* ***measure*** *it."*
2. **The gap:** CAG — only 31% of cities can *act* on their own data. Data everywhere, intelligence nowhere.
3. **Reveal the blame map (Innovation):** click a ward → *"68% construction dust, confidence 0.83"* — satellite + sensors agree. Toggle the **satellite-vision layer** (the exact construction sites our CV model detected) and switch from sparse station-dots to the **dense 1km grid**. *"We don't measure pollution. We assign blame."*

4. **Forecast (Technical proof):** scrub 72h → a spike tomorrow 6 PM; show the slide: **"25%+ better than persistence — here's the backtest."**
5. **Action (Business impact):** enforcement worklist → top-3 construction sites = 41% of the ward's PM2.5 → auto-generated, CPCB-cited notice. The dossier even **shows the Sentinel-2 image** of the plume beside the cited rule (E6). *Signal → action in under 5 minutes.* 5b. **What-if (prescriptive):** officer toggles *"pause construction, Ward 12"* → the map **re-forecasts live** and AQI drops 'Very Poor' → 'Moderate'. *"We don't just predict — we prescribe."* 5c. **Optimise (the closer):** *"optimise for tomorrow's spike under 20 inspector-hours"* → VayuNetra returns the best action bundle — *"−18 AQI · 42k people protected · ≈ ₹0.6 cr health cost avoided · ~9 t CO_2e"* (E5 + E7). *Numbers illustrative; each cites its factor.*
6. **Protect citizens (UX):** a judge **messages the live Telegram bot** (or calls the pre-verified IVR number) and gets a real Kannada advisory; phone shows the ward alert.
7. **Scale (Scalability):** switch Delhi → Bengaluru → Mumbai; onboard a 4th city from config live. *"Same engine. 131 NCAP cities."*
8. **Close (impact + ask):** the latency we collapsed, the lives at stake, the path to national scale.

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20. Business Impact, Market & Go-to-Market

20.1 Impact thesis (India-scale, the 25% Business-Impact axis)

- **Health:** air pollution → **1.67M premature deaths/yr**; reducing source exposure even marginally has enormous QALY value.
- **Operational:** collapses signal-to-action from **weeks/never (CAG)** → **minutes**; turns scarce inspector capacity into a **targeted**, exposure-maximising resource.
- **Governance:** gives **131 NCAP cities** an off-the-shelf intelligence layer they currently lack.
- **Quantified (E7):** every intervention is costed in **₹ health savings · respiratory cases prevented · CO_2e co-benefit** — concrete rupee impact for the business case, plus a Sustainability / net-zero bridge.

20.2 Market & buyers

- **Primary:** State Pollution Control Boards, Municipal Corporations, Smart City SPVs, CPCB.
- **Funding rails:** NCAP allocations + 15th Finance Commission air-quality grants for million-plus cities + Smart Cities Mission + CSR/ESG. *(Exact figures to be cited from latest budget docs before the pitch — see Appendix assumptions.)*
- **Adjacent:** public-health agencies, real-estate/ESG reporting, insurers, schools/hospitals.

20.3 Go-to-market (post-hackathon vision)

1. Land 1–2 lighthouse cities (likely a Smart City SPV) as a paid pilot.
2. Prove an intervention-effectiveness number (PM2.5 ↓ in targeted wards).
3. Expand city-by-city via the config-driven engine; SaaS + per-city licensing; CPCB national tier.

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21. Appendix

21.1 Glossary

- **AQI** — Air Quality Index. **CAAQMS** — Continuous Ambient Air Quality Monitoring Station. **NCAP** — National Clean Air Programme. **GRAP** — Graded Response Action Plan. **SPCB/CPCB** — State/Central Pollution Control Board. **AOD** — Aerosol Optical Depth. **H3** — Uber's hexagonal geospatial grid. **RAG** — Retrieval-

Augmented Generation. **BLH** — Boundary-Layer Height. **TROPOMI** — sensor on Sentinel-5P. **HYSPLIT** — atmospheric trajectory/dispersion model. **TFT/GNN** — Temporal Fusion Transformer / Graph Neural Network. **SAFAR/TERI** — Indian air-quality research/inventory sources.

21.2 Metric definitions

- **Persistence baseline:** forecast(t+h) = value(t). The brief grades forecast skill against this.
- **Forecast skill score:** `1 − RMSE_model / RMSE_persistence` (>0 means better than persistence).
- **Attribution agreement:** |our_share − inventory_share| per category, averaged.
- **Signal-to-action latency:** wall-clock from new signal ingestion → enforcement recommendation emitted.

21.3 Key data links (to wire up)

- CPCB CAAQMS / data.gov.in · OpenAQ API · Google Earth Engine (Sentinel-5P/2, MODIS/VIIRS) · IMD / OpenWeather / ERA5 / GFS · OpenStreetMap · WorldPop · SAFAR / TERI inventories · NCAP & GRAP documents.

21.4 Figures: provenance & assumptions

- **From the official PS5 brief (firm):** Delhi AQI 218 / 200+ days; Mumbai 60+ days; Kolkata >150; 24 of top-50 cities Tier-1/2; **1.67M** premature deaths (Lancet); **900+** CAAQMS; **2024 CAG audit — only 31%** with actionable protocols.
- **Supporting figures to verify before final pitch:** NCAP "131 non-attainment cities", 15th Finance Commission air-quality grant amounts, pollution % of GDP. *Cite from primary sources in the deck; do not present unverified numbers to judges.*

21.5 Open questions for the team

Resolved (locked): ✅ cities = Delhi + Bengaluru + Mumbai (languages hi/en/kn/mr) · ✅ infra = 100% free-tier (₹0) · ✅ no mandated sponsor tools · ✅ no domain expert → CPCB/GRAP rubric proxy.

Still open:

1. Confirm product **name** (default **VayuNetra**) for repo + branding before scaffolding.
2. **Forecast finale model** — GNN vs Temporal Fusion Transformer: decide after the MVP backtest (pick whichever beats persistence most on held-out data).
3. **IVR for the demo** — Twilio trial calls reach only *verified* numbers, so either (a) verify the judges'/demo number in advance, or (b) use an in-app call simulation; **Telegram bot is the always-free default** regardless.
4. **Repo visibility** — make it **public** to get unlimited free GitHub Actions minutes (also an open-source pitch point).

21.6 Key differentiators vs competition (the "Why #1" cheat-sheet)

Most teams will build…	VayuNetra delivers…
An AQI **dashboard** (measure)	An **action engine** — attribution → forecast → enforcement → advisory
A single LLM / one agent	A genuine **multi-agent** system + geospatial + RAG + **physics-informed ML**
"AI forecast" (no baseline)	A forecast **provably beating persistence**, reproducible live on stage
Generic source guesses	**Satellite-ground blame map** with confidence + SHAP + **CV-detected sources**
Predictive only	**Prescriptive what-if** — *"≈18k people protected, 2.3 t PM2.5 avoided"*
One hardcoded city	**City-agnostic** — 3 live + a 4th onboarded on stage, at **₹0** infra

Most teams will build…	VayuNetra delivers…
A fragile live demo	**Demo Mode** — runs 100% offline, *cannot* break
Hand-waves on bias	A **quantified fairness audit** (priority driven by pollution, not poverty)
Predicts (maybe one what-if)	A **prescriptive optimiser** — the best intervention *bundle* under an inspector budget
Text-only citations	**Multimodal dossiers** — cite the actual satellite image of the violation
Reports AQI deltas	Impact in **₹ health savings · cases prevented · CO_2e**

End of PRD v1.4 — built to win PS5 at ET AI Hackathon 2026. Companion: ARCHITECTURE.md (v1.4, in sync). Next artifacts on request: data-source integration runbook, validation/backtesting notebook plan, pitch-deck outline, and a P0 starter scaffold.

PART II — TECHNICAL ARCHITECTURE

VayuNetra — Technical Architecture Specification

> Companion to **PRD.md**. This document is the **buildable blueprint** for a 2–3 person team. It specifies components, data models, agent contracts, APIs, the ML/RAG subsystems, deployment, and — critically — a **100% free-tier ($0) implementation** path.
>
> **Project:** VayuNetra · **Problem Statement:** PS5 — AI-Powered Urban Air Quality Intelligence · **Hackathon:** ET AI Hackathon 2026 · **Spec version:** v1.4 (synced to PRD)

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1. Purpose & Architectural Constraints

This spec turns the PRD into something the team can build against tomorrow.

Hard constraints (from team decisions):

Constraint	Implication
$0 / free-tier only	Every component must have a genuine free tier; no paid infra. Documented limits + mitigations.
City-agnostic, multi-city from day one	Universal spatial key (H3) + config-driven city onboarding; **no per-city code**.
Showcase: Delhi + Bengaluru + Mumbai	Languages: **Hindi, English, Kannada, Marathi**.
No domain expert	Enforcement quality validated by a transparent **CPCB/GRAP-derived rubric proxy**, not claimed expert review.
2–3 person team, time not constrained	Favor managed free services over self-hosting; minimize ops burden.
Multi-agent + geospatial + RAG	The architecture must visibly embody all three (the organizers' signal).

Bonus the $0 constraint buys us: "VayuNetra runs entirely on free and open infrastructure — any of India's 131 NCAP cities can adopt it at near-zero cost." This is a pitch asset (Scalability + Business Impact).

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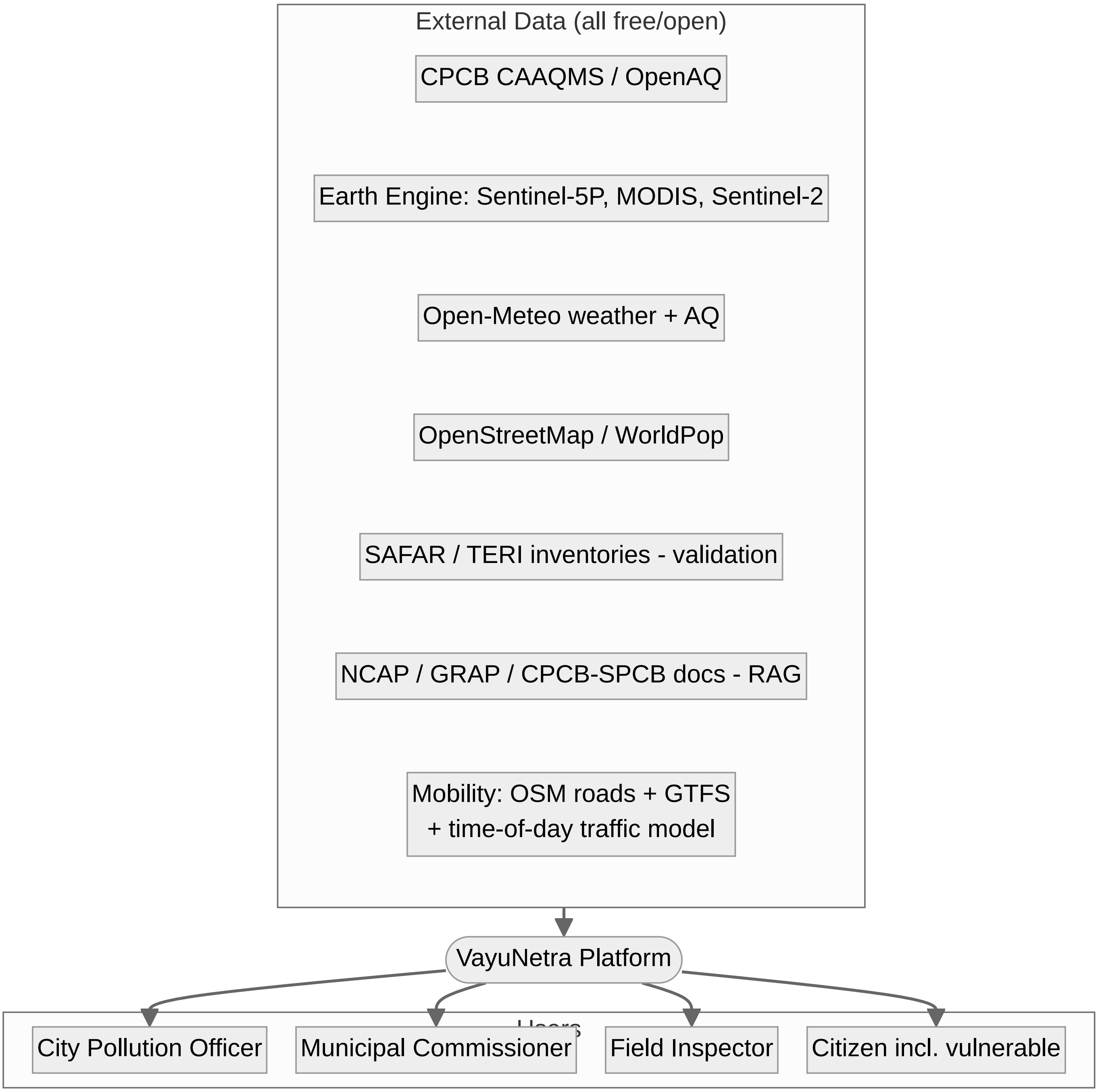
2. Guiding Principles

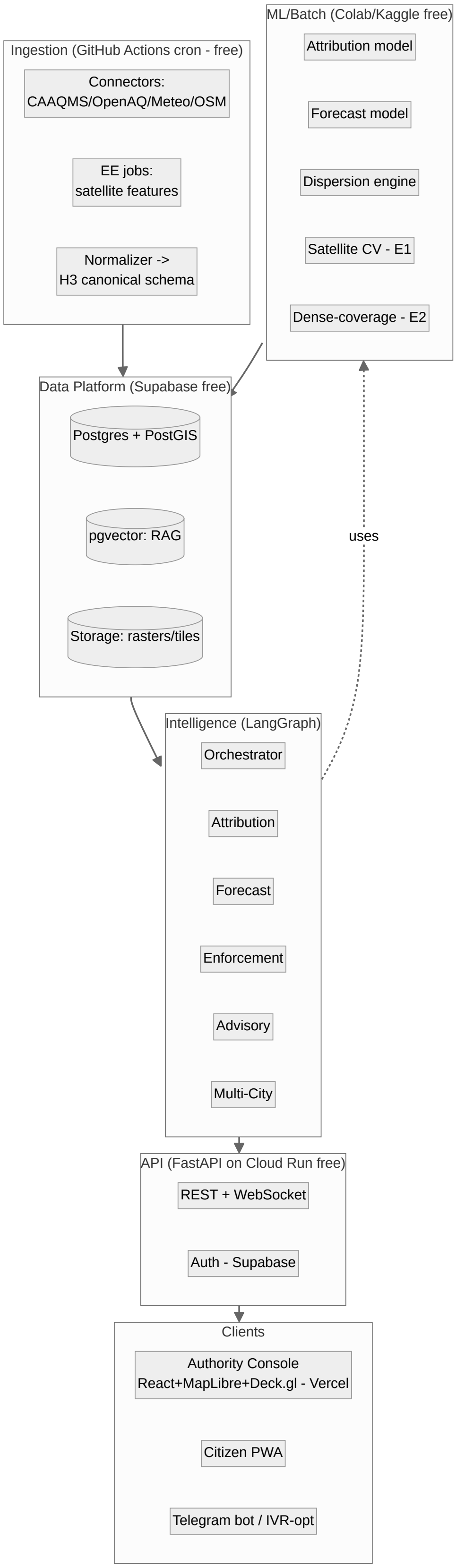
1. **Action over measurement** — every component exists to produce an *action* (attribution → forecast → enforcement → advisory), never just a number.
2. **City-agnostic by construction** — all data normalizes to `(city_id, h3_cell, ts, variable, value, source, confidence)`.
3. **Physics + ML hybrid** — dispersion + chemical priors keep ML explainable and legally defensible.
4. **Everything is cited & traced** — attribution confidence, RAG citations, agent traces. Judges trust what they can audit.
5. **Free, but not fragile** — cache aggressively, degrade gracefully, never let a flaky live API break the demo.
6. **Model-agnostic LLM** — swap Gemini/Groq/Ollama via config; if the hackathon grants LLM credits, switch with one env var.

---

3. System Context (C4 L1)



4. Container / Component View (C4 L2)



5. The Free-First Technology Stack ($0)

Layer	Choice	Free tier / why	Limit & mitigation
Satellite / RS	**Google Earth Engine**	Free for research/non-commercial (hackathon qualifies); Google runs the compute	Quotas → precompute city features nightly, cache in Storage. Backup: **MS Planetary Computer** (free)
Weather + AQ forecast feed	**Open-Meteo**	Free, **no API key**, includes forecast + Air-Quality API	Rate limits generous; cache hourly
Ground AQI	**CPCB (data.gov.in) + OpenAQ**	Free API key / free API	CPCB flaky → OpenAQ + historical backfill
Mobility feeds (PS5-named)	OSM roads + **GTFS** transit + time-of-day traffic model	Free	real-time traffic is paid → engineered proxy (known limitation, §24)
Database	**Supabase** (Postgres 15)	Free: **PostGIS + pgvector + Auth + Realtime**, 500 MB DB, 1 GB storage	500 MB → store recent+aggregated; archive rasters to Storage/R2
Object storage	Supabase Storage / **Cloudflare R2** free	Rasters, tiles, model artifacts	R2 10 GB free egress-free
Vector store (RAG)	**pgvector** on Supabase	No extra service	Small corpus; fine on free tier
Agent framework	**LangGraph** (OSS)	Free, stateful multi-agent, traceable	—
LLM (reasoning/RAG/i18n)	**Google Gemini API free tier** (Flash)	Free, strong **multilingual**, good reasoning	Rate limits → cache + batch. Fallbacks: **Groq** (free Llama 3.3), **Ollama** (local)
Embeddings	**Default: local `bge-small`** via sentence-transformers (zero-API, no rate limits); Gemini text-embedding optional	Free	Local default avoids API throttling
ML training	**Google Colab** (free T4) + **Kaggle** (30h/wk GPU)	Free GPU for forecast/attribution models	Session limits → checkpoint to Storage
ML libs	LightGBM/XGBoost (MVP), PyTorch + torchvision / segmentation-models-pytorch (satellite CV, GNN/TFT), scikit-learn	OSS	—
Dispersion	Custom **Gaussian plume** (Python) + **HYSPLIT** (NOAA, free)	OSS/free	Simplify to plume + wind advection if time-boxed
Backend / API	**FastAPI** on **Google Cloud Run** free tier (2M req/mo, scale-to-zero)	Free, persistent URL	Cold starts → min-instance off; warm before demo. Alt: **Render/HF Spaces**
Pipelines / scheduler	**GitHub Actions** cron (2000 min/mo free)	Free scheduled data refresh + forecast runs	Keep jobs < keep within minutes; stagger
Frontend hosting	**Vercel** / **Netlify** free	Free CDN + CI	—
Maps	**MapLibre GL** + **Deck.gl** (`H3HexagonLayer`) + **Protomaps/MapTiler free** basemap	Fully free (no Mapbox token)	MapTiler free key or self-host Protomaps PMTiles

Layer	Choice	Free tier / why	Limit & mitigation
Citizen channels	**Telegram Bot API** (100% free) + **PWA**; **Twilio trial** for IVR demo	Telegram free forever; Twilio trial credit for a callable number	WhatsApp Business has cost → Telegram primary, WhatsApp "production upgrade"
Auth	**Supabase Auth**	Free, role-based	—
Observability	Structured logs + in-app latency widget; **Grafana Cloud** free (optional)	Free	—
CI/CD + Repo	**GitHub** + Actions	Free	—

> **Net infra cost target: ₹0.** Document any component that risks crossing a free limit in §22.

> **Model/cloud-agnostic:** the PRD stack rows are synced to this free-first design. The LLM and cloud are swappable via config — if the hackathon later grants Claude or GCP/AWS credits, switching is a one-env-var change with zero architectural impact.

6. Spatial Data Model (H3)

- **Universal spatial key = Uber H3.** Primary resolution **res 8** (avg hexagon ≈ **0.74 km²**, edge ≈ 0.46 km) → satisfies the brief's "~1 km grid."
- **Hierarchy:** res 6 (city zones) ⊃ res 8 (hyperlocal cells) ⊃ res 9 (fine, optional). Ward boundaries mapped to covering H3 sets.
- **Why H3:** (1) one math for every city (city-agnostic), (2) clean spatial joins across sensors/satellite/population/sources, (3) native Deck.gl `H3HexagonLayer` rendering, (4) cheap aggregation up/down resolutions.
- **Ward ↔ H3 mapping table** lets us present results at administrative ward level (officer-friendly) while computing on H3 (uniform).

7. Data Architecture

7.1 Connectors → normalization

Each external source has a connector that maps raw payloads to the **canonical measurement** record. Connectors are city-agnostic; a city is just a config (bbox, CAAQMS station ids, ward GeoJSON, language set).

Connectors: CAAQMS · OpenAQ · Open-Meteo · Earth Engine (satellite) · OSM · **GTFS / mobility** · **WorldPop / population-vulnerability** · SAFAR/TERI (validation) · NCAP/GRAP docs (RAG).

7.2 Canonical schemas (Postgres + PostGIS on Supabase)

```
-- Cities = config-driven onboarding (NO per-city code)
cities(
  city_id text primary key, name text, state text,
  bbox geometry(Polygon,4326), center geometry(Point,4326),
  languages text[],            -- e.g. {hi,en,kn,mr}
  caaqms_station_ids text[], ward_geojson_ref text,
  active boolean default true
```

```
);

-- Universal measurements (ground + satellite + weather, all here)
measurements(
  id bigserial primary key, city_id text, h3_cell text,         -- res 8
  station_id text null, ts timestamptz,
  variable text,               -- pm25,pm10,no2,so2,co,o3,aod,fire,wind_u,wind_v,blh,...
  value double precision, unit text,
  source text,                 -- caaqms,openaq,s5p,modis,s2,openmeteo
  confidence double precision default 1.0, ingested_at timestamptz default now()
);
create index on measurements(city_id, variable, ts);
create index on measurements(h3_cell, ts);

-- Source attribution (Agent 1 output) — the "blame map"
attribution(
  city_id text, h3_cell text, ts_window tstzrange,
  source_category text,        --
traffic,construction_dust,industrial,biomass_burning,transported,other
  share double precision,      -- 0..1, sums to 1 per cell/window
  confidence double precision, method_version text,
  evidence jsonb               -- which signals drove it (for explainability)
);

-- Forecasts (Agent 2 output) — incl. baseline for honest comparison
forecasts(
  city_id text, h3_cell text, issued_at timestamptz,
  horizon_h int,               -- 24,48,72
  target_var text default 'aqi',
  value double precision, pi_low double precision, pi_high double precision,
  persistence_value double precision,    -- baseline shown side-by-side
  model_version text
);

-- Emission source registry (Agent 3 inputs; CV-detected sources from E1 land here too)
emission_sources(
  id bigserial primary key, city_id text, geom geometry(Geometry,4326),
  type text,                   -- industry,construction,waste_burn,diesel_corridor
  name text, registry_ref text,
  source_origin text default 'registry',   -- registry | cv_detected (E1)
  detection_confidence double precision,    -- for CV-detected sources
  attributes jsonb
);

-- Enforcement recommendations (Agent 3 output)
enforcement_recs(
  id bigserial primary key, city_id text, h3_cell text, ts timestamptz,
  source_id bigint null, priority_score double precision,
  contribution double precision, pop_exposed int,
  rationale text, evidence jsonb, rag_citations jsonb,
  rubric_score jsonb,          -- §14 proxy score
  status text default 'proposed'
);

-- Citizen advisories (Agent 4 output)
advisories(
  id bigserial primary key, city_id text, ward_id text, h3_cell text,
  issued_at timestamptz, horizon_h int, risk_tier text,
  audience_segment text,       -- general,outdoor_worker,elderly,school,respiratory
```

```
  language text, channel text, message text
);

-- RAG knowledge base (pgvector) — text + (E6) multimodal satellite image-patch embeddings
kb_chunks(
  id bigserial primary key, doc_id text, title text, source_url text,
  modality text default 'text',     -- text | image (E6 Sentinel-2 patch)
  chunk_text text, image_ref text,  -- image_ref -> Storage/R2 patch when modality='image'
  embedding vector(768), metadata jsonb
);
create index on kb_chunks using ivfflat (embedding vector_cosine_ops);

-- Latency telemetry (proves North-Star metric)
action_traces(
  id bigserial primary key, city_id text, signal_ts timestamptz,
  attribution_ts timestamptz, forecast_ts timestamptz,
  enforcement_ts timestamptz, advisory_ts timestamptz,
  total_latency_ms int, trace jsonb
);
```

7.3 Storage tiers (free-tier sizing)

- **Hot (Postgres):** recent measurements (e.g., 30–90 days), latest attribution/forecast, registries, RAG. Keep < 500 MB by retaining recent + aggregated only.
- **Warm/Cold (Storage/R2):** raster tiles, satellite-derived feature grids, historical archives, model artifacts.
- **Aggregation job:** nightly roll-up of old fine-grain rows → daily/ward summaries; purge raw beyond retention.

7.4 Refresh pipelines (GitHub Actions cron — free)

Job	Schedule	Output
`ingest_ground`	hourly	CAAQMS/OpenAQ → measurements
`ingest_weather`	hourly	Open-Meteo forecast + AQ → measurements
`ee_satellite_features`	daily	Earth Engine → city feature grids → Storage + measurements
`run_attribution`	hourly	Agent 1 → attribution table
`run_forecast`	6-hourly	Agent 2 → forecasts (+ persistence)
`refresh_enforcement`	6-hourly	Agent 3 → enforcement_recs
`rollup_archive`	nightly	aggregate + purge to stay in free limits

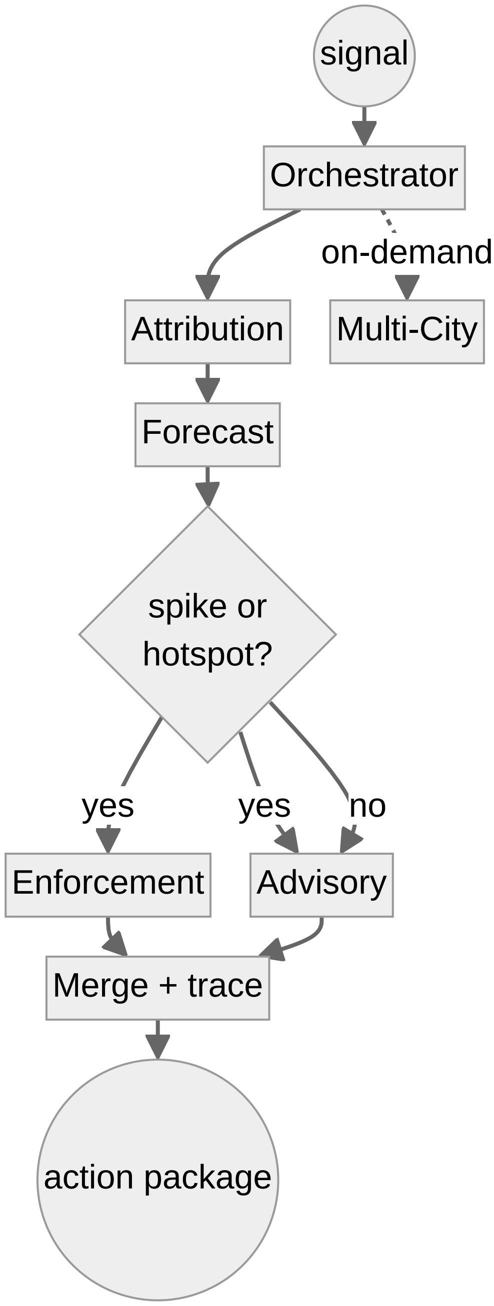
8. Multi-Agent Intelligence Layer (LangGraph)

8.1 Shared state (typed)

```
class GraphState(TypedDict):
    city_id: str
    time_window: tuple          # (start, end)
    focus_cells: list[str]      # H3 cells of interest (e.g., spiking)
    signals: dict               # latest measurements snapshot
    attribution: dict           # Agent 1 result
    forecast: dict              # Agent 2 result
    enforcement: list           # Agent 3 result
```

```
    advisories: list             # Agent 4 result
    comparison: dict             # Agent 5 result
    citations: list              # RAG sources used
    trace: list                  # per-node timing + decisions
    latency_ms: int
```

8.2 Graph topology



8.3 Agent contracts

Agent	Input	Output	Tools	LLM use
Orchestrator	trigger (signal/query/schedule)	routed plan + merged action package	state mgr, router	Light — routing/explanation only
A1 Attribution	measurements + satellite + land use (focus cells)	`attribution[]` + confidence + evidence	ML model svc, dispersion svc, SQL	Optional — narrate evidence

Agent	Input	Output	Tools	LLM use
A2 Forecast	met forecast + lags + dispersion + calendars	`forecasts[]` + intervals + persistence	ML model svc, Open-Meteo	None (pure ML)
A3 Enforcement	attribution + forecast + source registry + exposure	ranked `enforcement_recs[]` + dossiers	RAG, SQL, scorer	Yes — dossier synthesis w/ citations
A4 Advisory	forecast + vulnerability + health breakpoints	`advisories[]` multi-language	RAG (health), i18n, channels	Yes — localization/translation
A5 Multi-City	cross-city history + interventions	comparison + playbook recs	SQL, analytics	Yes — narrative synthesis

Design rule: ML/physics produce the *numbers*; the LLM only *explains, cites, localizes, and synthesizes*. This keeps outputs accurate and auditable (no hallucinated AQI).

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9. ML Subsystem

9.1 Attribution model (Agent 1)

- **Approach:** physics-informed features + supervised apportionment.
- **Features per H3 cell/time:** pollutant ratios (PM10:PM2.5, NO_2, SO_2, CO, O_3), satellite NO_2 column, AOD, fire-pixel proximity, land-use fractions (industrial/road/built), traffic proxy, wind-advected upwind signals, dispersion output.
- **Model:** gradient boosting (multi-output) → source-share vector; calibrated; **SHAP** for explainability ("why this blame").
- **Labels/calibration:** anchor to published SAFAR/TERI apportionment for validation wards (held out, never trained on).

9.2 Forecast model (Agent 2)

- **Target:** AQI / PM2.5 per H3 cell at +24/+48/+72h.
- **Features:** met forecasts (wind u/v, BLH, temp, RH, precip from Open-Meteo), lagged AQI, dispersion prior, traffic + **seasonal/event calendars** (stubble windows, Diwali, winter inversion), spatial neighbors, hour/day-of-week.
- **Models: MVP = LightGBM** w/ quantile loss (intervals). **Finale = spatiotemporal GNN / Temporal Fusion Transformer** (PyTorch, trained on Colab/Kaggle).
- **Baselines (mandatory): persistence** (`ŷ(t+h)=y(t)`) + climatology. Stored alongside every forecast for side-by-side proof.
- **Serving:** batch (GitHub Actions) writes `forecasts`; API reads. Lightweight model can also run in-API for live "what-if."
- **Physics-informed:** the dispersion output (§9.3) enters as a feature/constraint → a genuine **physics-informed ML hybrid** (explainable + atmospheric-science-grounded), the "novel ML" depth judges want without black-box risk. *Optional stretch (not committed):* full **PINN** (advection-diffusion in the loss) — high upside, high risk; core-first only.

9.3 Dispersion engine (physics prior)

- **Gaussian plume** for local point/area sources (industry/construction) + **wind-field advection** of satellite NO_2/AOD for transported pollution; optional **HYSPLIT** back-trajectories.

- Feeds attribution (transported share) and forecast (physics feature). Pure Python; runs in batch.

9.4 Model ops (free)

- Versioned artifacts in Storage/R2; metadata in Postgres; experiment tracking via lightweight logs or **MLflow** (local/HF). No paid registry.

9.5 Satellite source-detection CV — E1 (trained)

- **Model:** semantic segmentation / detection (U-Net or Mask R-CNN with a pretrained encoder) on **Sentinel-2** (10 m bands) + optional **Sentinel-1 SAR** for cloud robustness; classes = construction, brick kiln, open burning.
- **Labels:** bootstrap from OSM land use + known sites + a small hand-labelled tile set; transfer-learning keeps the labelled set small.
- **Output:** geo-located detections → `emission_sources` (auto-populates enforcement). Trained on Kaggle GPU; inferred in the daily EE job.

9.6 Dense-coverage models — E2 (trained)

- **AOD → PM2.5:** GBM/MLP regressor on paired satellite AOD + met → surface PM2.5; fills no-station areas.
- **1km downscaling:** lightweight CNN / learned super-resolution fusing station + satellite + land use → dense H3-res-8 field + uncertainty. Turns ~40 stations into a full-city 1 km map.

9.7 Intervention what-if simulator — E3

- **Engine:** counterfactual dispersion + forecast run with a source toggled off/reduced → ΔAQI per H3 cell; served via `POST /simulate`; powers the prescriptive demo. Reuses §9.2 + §9.3 (no new model to train).
- **Impact quantification:** ΔAQI × WorldPop population × exposure-response → **people protected + PM2.5 tonnes avoided + exposure-hours reduced** per scenario (returned by `/simulate`; the Business-Impact numbers on the demo card).

9.8 Spike / anomaly detector — E4 (trained, stretch)

- **Model:** STL decomposition + isolation forest / autoencoder over per-cell series; flags novel events → enforcement queue.

9.9 Training plan & compute (free)

- **Where:** Google Colab (free T4) + Kaggle (30 h/wk GPU). **Tracking:** MLflow-lite + metrics in Postgres. **Artifacts:** versioned to R2/Storage. **Reproducibility:** one `evaluate.ipynb` regenerates every metric (no leakage, fixed seeds, SAFAR/TERI held out).
- **Serving:** batch inference in GitHub Actions writes results to Supabase; light models run in-API for live what-if.

9.10 Responsible AI & fairness

- **Equity guard:** enforcement prioritisation is monitored so it does **not** systematically over-target low-income wards; **human approval** is required before any action.
- **Privacy:** citizen advisory uses **ward-level**, not individual, location by default. Every model output carries a confidence score.

9.11 Prescriptive optimiser — E5

- **Engine:** greedy / priority-knapsack over candidate interventions (construction halts, corridor diversions, CV-detected-site inspections); each scored via the E3 simulator by exposure-weighted ΔAQI ÷ resource cost, under an inspector-hour / budget constraint. Returns top-3 ranked packages. Served via `POST /optimize`. **No training** — search over existing simulators.

9.12 Multimodal RAG — satellite visual evidence — E6

- **Engine:** a free **CLIP** / vision encoder embeds Sentinel-2 patches → pgvector (`kb_chunks.modality='image'`, `image_ref` → Storage/R2). The enforcement dossier retrieves the most relevant patch **+** governing rule, citing both. Pretrained encoder; **no custom training**.

9.13 Health & carbon quantification — E7

- **Engine:** static, **citable** factor tables — WHO/CPCB dose-response (ΔPM2.5 → cases / mortality risk → ₹ health cost) + emission factors (→ CO_2e). Applied across attribution / forecast / what-if / optimiser; **every figure cites its source**. Feeds `/simulate`, `/optimize`, and advisory cards.

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10. RAG Subsystem

- **Corpus:** NCAP, **GRAP** action matrices, CPCB/SPCB regulations & consent norms, CPCB/WHO AQI **health breakpoints**, source-apportionment literature, enforcement SOPs.
- **Ingestion:** PDF/HTML → clean text → semantic chunking → embeddings (**default local `bge-small`**, Gemini optional) → `kb_chunks` (pgvector).
- **Retrieval:** cosine top-k + metadata filters (city/topic) → LLM synthesizes **cited** answers (every claim links to a source chunk).
- **Used by:** A3 (regulatory basis for enforcement dossiers) and A4 (health-threshold grounding for advisories).
- **Why it matters:** citations make enforcement **court-defensible** and make judges trust the system.
- **Multimodal (E6):** Sentinel-2 image patches are CLIP-embedded into the same pgvector store (`modality='image'`); enforcement dossiers retrieve and **cite the visual evidence** alongside the governing rule — the officer sees the satellite proof, not just text.

---

11. API Design

FastAPI (async) on Cloud Run free tier. Auth via Supabase JWT; role-gated.

Method	Endpoint	Purpose	Role
GET	`/cities`	list onboarded cities	all
GET	`/aqi/current?city&bbox`	live AQI per H3	all
GET	`/attribution?city&cell|ward&ts`	source split + confidence	officer+
GET	`/forecast?city&cell&horizon`	forecast + intervals + persistence	all
GET	`/enforcement?city&date`	ranked recommendations	officer+
POST	`/enforcement/{id}/dossier`	generate cited evidence packet + **satellite visual evidence** (E6)	officer+
GET	`/advisory?city&ward&lang`	localized citizen advisory	all
POST	`/agent/query`	conversational orchestrator (NL → action)	officer+
POST	`/simulate`	what-if intervention → ΔAQI + people protected / PM2.5 avoided + ₹/CO_2e (E3, E7)	officer+

Method	Endpoint	Purpose	Role
POST	`/optimize`	best intervention bundle under a resource/inspector budget → top-3 packages (E5)	officer+
POST	`/admin/cities`	**onboard a city via config** (scalability demo)	admin
WS	`/live`	push attribution/forecast/alert updates	all

Envelope (per global API standard): `{ success, data, error, meta }`.

---

12. Frontend Architecture

Authority Console — React + TypeScript + Vite + Tailwind + **MapLibre GL** + **Deck.gl** (Vercel).

- **Map-first** layout; default = **Blame Map** (`H3HexagonLayer` colored by *dominant source*, not just AQI).
- **Layer panel:** live AQI · source attribution · 24–72h forecast (time-slider) · enforcement targets · population vulnerability · satellite overlays.
- **Key demo toggles: "Detected Sources"** (E1 satellite-CV construction/kiln/burn) · **"Stations-only ↔ Dense 1km grid"** (E2) · **"What-if intervention"** (E3, calls `POST /simulate`) · **SHAP tooltips** on the blame map (*"NO_2 + AOD drove 68% construction"*).
- **Action rail (right):** ranked enforcement worklist → click → cited dossier → **"Generate Cited Dossier (PDF export)"** + "Dispatch / Generate Notice".
- **Top bar: city switcher** (multi-city proof), time scrubber ("now" vs "tomorrow 18:00"), comparative tab, and a live **"Signal → Action: 2m 47s" latency widget**.
- **Fairness panel:** enforcement-action distribution across socio-economic wards (pre-empts any bias concern).
- **State:** TanStack Query (server cache) + Zustand (UI state); WebSocket for live updates.

Citizen PWA — installable, offline-capable, ward AQI + 72h + personalized health action, language toggle (hi/en/kn/mr).

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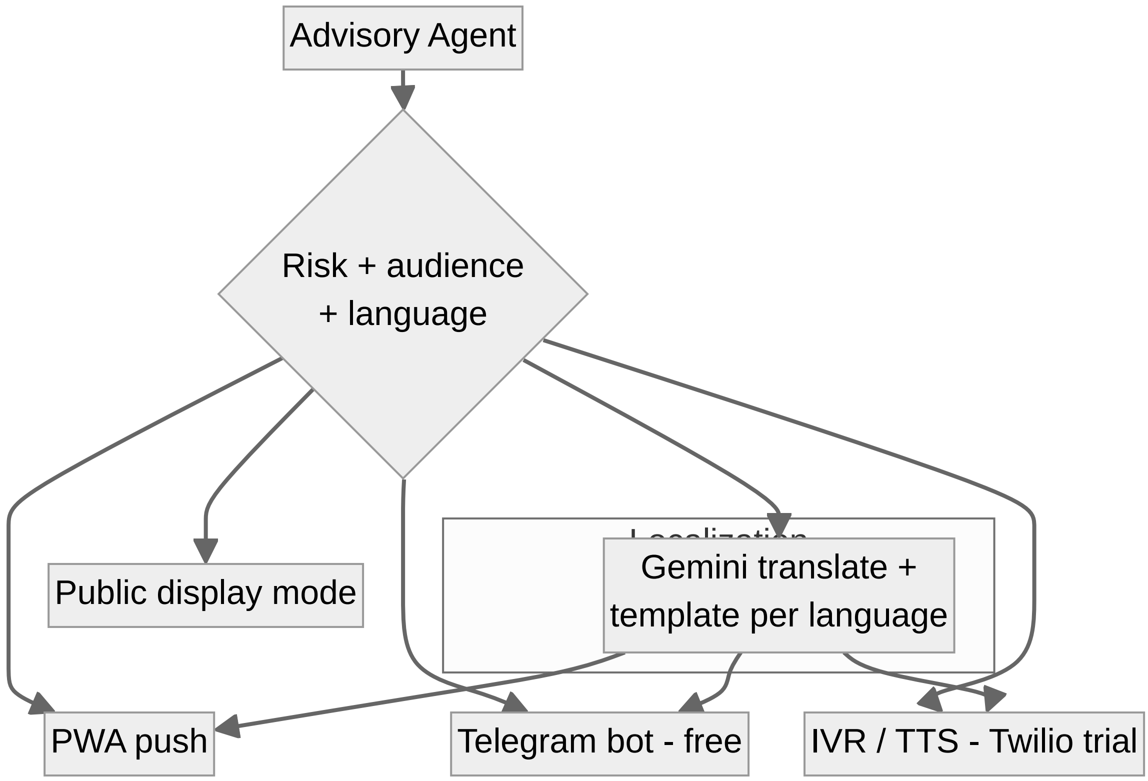
13. Citizen Channel Architecture



- **Primary (free):** Telegram bot + PWA. **IVR demo:** Twilio trial number (judges can call, hear Kannada/Marathi TTS). **WhatsApp** = production upgrade (has cost).
- **Localization:** short templated messages + LLM translation, **native-speaker reviewed** for the 4 demo languages to guarantee quality.

---

14. Validation & Evaluation Harness

> Treats the brief's **Evaluation Focus** as a test suite. Designed for **no domain expert** (per your decision) — enforcement quality uses a transparent, authoritative **rubric proxy**.

#	Validates	Method	Output artifact
1	**Attribution accuracy**	Compare ward apportionment to **held-out SAFAR/TERI**	agreement table + map overlay ("within ±X%/category")
2	**Forecast skill**	Temporal-split backtest; RMSE/MAE @24/48/72h vs **persistence + climatology**	skill score `1 − RMSE_model/RMSE_persistence` ; target ≥0.25
3	**Enforcement quality (rubric proxy)**	Score top-10 recs on a **CPCB/GRAP-derived rubric** (below)	composite precision; transparent, no expert claim
4	**Advisory relevance & coverage**	Native-speaker review + readability; count languages	≥4 languages, relevance notes
5	**Signal-to-action latency**	Instrument `action_traces`	median ms; contrast with CAG status quo
6	**Satellite CV detection (E1)**	precision/recall on held-out labelled tiles	mAP / F1

#	Validates	Method	Output artifact
7	**Dense-coverage (E2)**	AOD→PM2.5 + downscaling RMSE at held-out stations vs interpolation	RMSE / skill score
8	**What-if simulator plausibility (E3)**	sanity vs dispersion physics + historical analogues	directional correctness + magnitude sanity
9	**Fairness / equity audit**	partial corr(enforcement priority, ward income | source contribution, exposure)	≈ 0 — income adds no independent signal (driven by pollution, not poverty)
10	**Optimiser quality (E5)**	optimiser package vs best-single-action + random baselines (simulated)	ΔAQI per inspector-hour improvement
11	**Multimodal retrieval (E6)**	CLIP patch retrieval relevance on held-out tiles	precision@k

Enforcement rubric proxy (0–2 each, transparent & defensible):

1. **Attribution match** — target's type matches the cell's dominant attributed source.
2. **Actionability** — maps to a real **GRAP/CPCB** action lever.
3. **Exposure impact** — population × contribution above threshold.
4. **Regulatory grounding** — a citable rule exists (RAG-retrieved).
5. **Confidence** — attribution confidence above threshold.

→ Composite ≥ 7/10 = "act". Report top-10 precision. **Framed honestly:** "rubric-based evaluation grounded in the CPCB GRAP action framework; independent expert validation is a planned next step."

Reproducibility: a single `evaluate.ipynb` (Colab) regenerates every number live for judges. Fixed seeds, no leakage, validation inventories never trained on.

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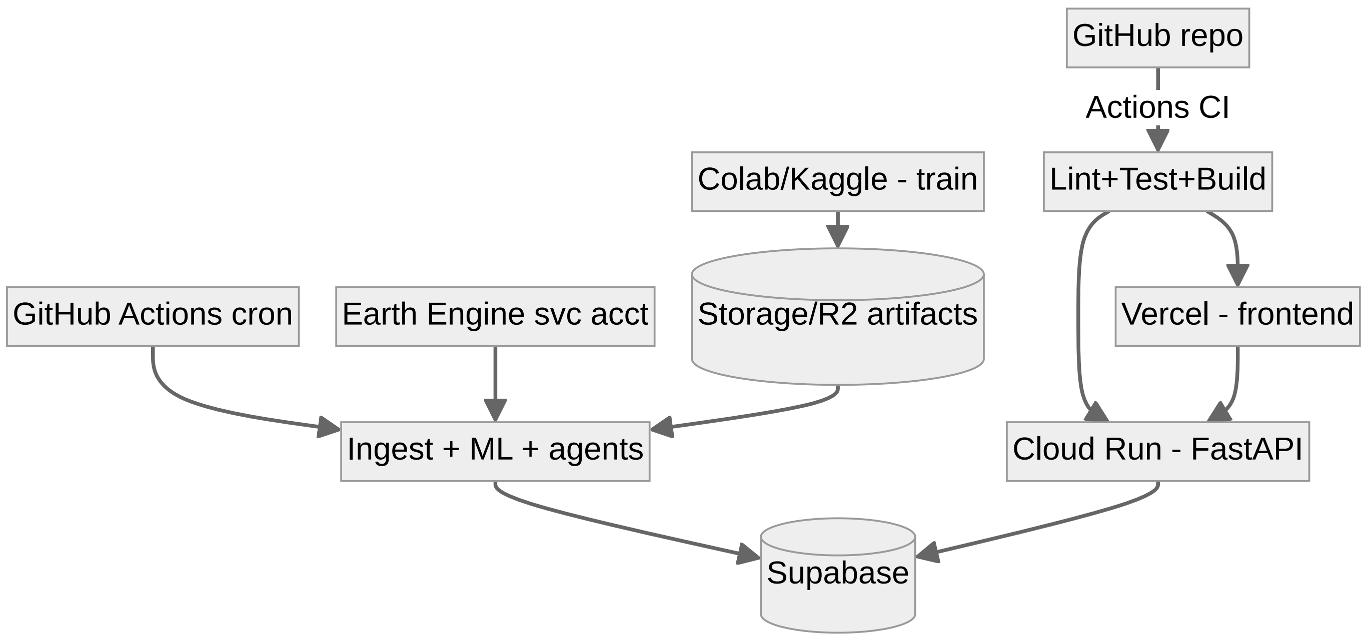
15. Deployment Topology & CI/CD



Concern	Choice (free)
Environments	`preview` (Vercel/PR) + `prod` (single, for demo)
Backend	Cloud Run (scale-to-zero, free tier)

Concern	Choice (free)
Frontend	Vercel
Scheduled compute	GitHub Actions cron
Secrets	GitHub Actions secrets + Cloud Run env + Supabase vault
Demo safety	**Demo Mode** (`DEMO_MODE=true`): a frozen, versioned, deterministic snapshot of all 3 cities (raw + precomputed agent outputs) runs the **entire scored demo offline** — zero live-API dependency; the live system is shown alongside. Plus pre-warm Cloud Run + OpenAQ fallback.

16. Security, Auth & Roles

- **Auth:** Supabase Auth (JWT). **Roles:** `admin` , `officer` , `inspector` , `citizen` (RLS policies in Postgres).
- **RLS:** citizens read only public AQI/advisory; officers read attribution/enforcement for their city; admin onboards cities.
- **Input validation:** schema-validate all API inputs (Pydantic) — never trust external feeds.
- **Secrets:** no keys in code; env/secret store only (aligns with global security rules).
- **Human-in-the-loop:** enforcement actions are *recommended*, approved by an officer — never auto-dispatched.
- **Audit:** `action_traces` + agent traces give a full, reviewable decision log.
- **Responsible AI / fairness:** monitor enforcement prioritisation so it does not systematically over-target low-income wards (§9.10); human approval required; ward-level (not individual) citizen location.

17. Observability & Signal-to-Action Instrumentation

- **The North-Star metric is a feature, not a log line.** Every signal that flows through the graph stamps `action_traces` (signal→attribution→forecast→enforcement→advisory) and computes `total_latency_ms` .
- **Live latency widget** on the console shows "signal → action: 2m 41s" during the demo — a visceral proof point.
- **Agent traces** (LangGraph) are surfaced in a debug drawer → judges can audit *how* a decision was made.
- Basic logs/metrics to stdout (Cloud Run) + optional Grafana Cloud free.

18. Scalability & Multi-City Mechanics

- **Onboard a city = POST `/admin/cities`** with `{name, bbox, ward_geojson, caaqms_station_ids, languages}` . Pipelines auto-pick it up; H3 + satellite + weather are global. **Zero code change.**
- **Demo the claim:** run Delhi + Bengaluru + Mumbai; then **onboard a 4th city live from config** on stage.
- **Parallelism:** per-city jobs are independent; GitHub Actions matrix runs them concurrently.
- **Free-tier ceiling:** ~3–5 cities comfortably on Supabase free (with rollups). Narrative: "same engine → 131 NCAP cities" with a trivial paid-tier bump.

19. Non-Functional Requirements

NFR	Target
Map interaction latency	< 2 s for layer toggle / city switch
Forecast freshness	≤ 6 h old
Attribution freshness	≤ 1 h old
Signal-to-action latency	< 5 min (North Star)
Availability (demo window)	99% (pre-warmed)
Demo resilience	`DEMO_MODE` frozen offline snapshot — zero live-API dependency
Cost	**₹0** (free tiers)
Reproducibility	one-click `evaluate.ipynb` regenerates all metrics
Accessibility	citizen PWA WCAG-AA, 4 languages

20. Repository Structure

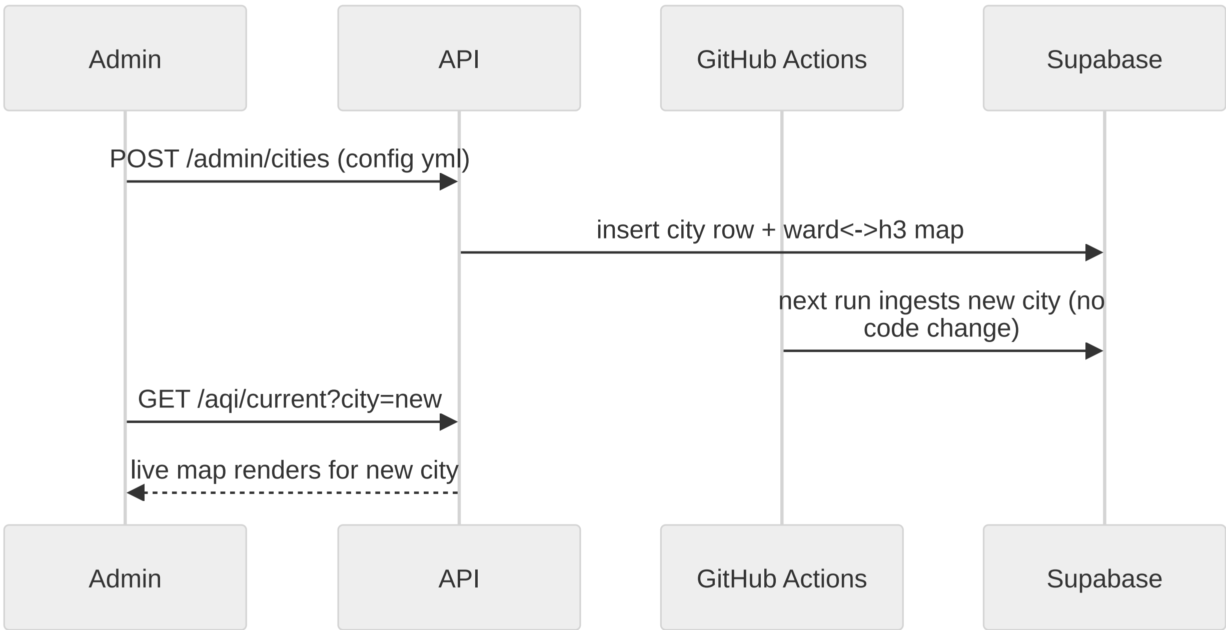
```
vayunetra/
├── README.md
├── docs/                  # PRD.md, ARCHITECTURE.md, diagrams, deck
├── infra/                 # GitHub Actions workflows, Cloud Run, supabase migrations
│   ├── workflows/         # ingest.yml, forecast.yml, attribution.yml, rollup.yml
│   └── supabase/          # SQL migrations (schemas in §7)
├── connectors/            # city-agnostic connectors (caaqms, openaq, openmeteo, ee, osm,
mobility/gtfs, worldpop)
├── core/
│   ├── spatial/           # H3 utils, ward<->h3 mapping
│   ├── schemas/           # pydantic models, canonical record
│   └── config/cities/     # delhi.yml, bengaluru.yml, mumbai.yml  <- onboarding = a file
├── ml/
│   ├── attribution/       # features, train, serve, SHAP
│   ├── forecast/          # features, baselines, lgbm, gnn, intervals
│   ├── dispersion/        # gaussian plume, advection, hysplit wrapper
│   ├── vision/            # E1: Sentinel-2 CV source detection + E6 CLIP patch embeddings
│   ├── coverage/          # E2: AOD->PM2.5 + 1km downscaling
│   ├── simulator/         # E3: what-if engine + E5 prescriptive optimiser
│   ├── impact/            # E7: health (dose-response) + carbon (emission-factor) quantification
│   └── anomaly/           # E4: spike/anomaly detector (stretch)
├── agents/                # LangGraph: orchestrator + 5 agents + tools
├── rag/                   # ingest, chunk, embed, retrieve
├── api/                   # FastAPI app, routers, auth, websocket
├── web/                   # React + MapLibre + Deck.gl console + citizen PWA
├── channels/              # telegram bot, IVR/TTS, i18n templates
├── eval/                  # evaluate.ipynb, backtests, rubric scorer
├── demo/                  # frozen deterministic snapshot for DEMO_MODE (offline, demo-proof)
└── tests/                 # unit + integration (TDD, 80%+ target)
```

Adding a city = drop a `core/config/cities/<city>.yml`. That's the scalability story in one folder.

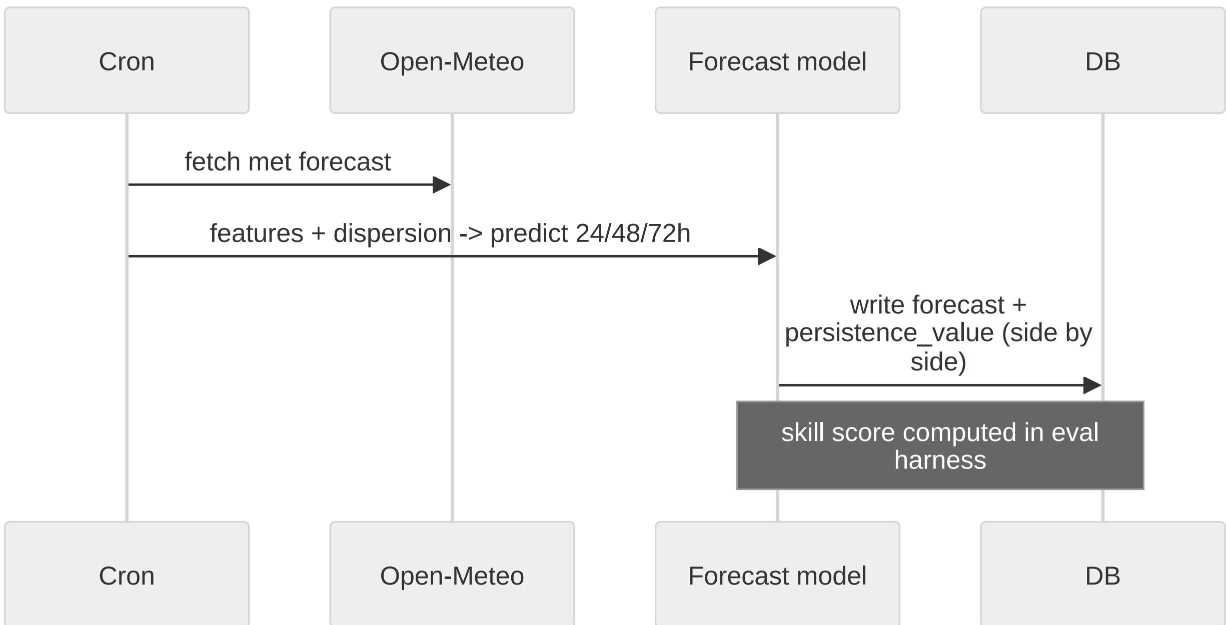
21. Key Sequence Flows

A. Live signal → action (the demo spine): see §8.2 graph + §17 latency stamping.

B. City onboarding (scalability proof):



C. Forecast run (baseline-honest):



22. Free-Tier Limits & Mitigations

> These are **operational envelopes, every one fully mitigated** — none affects the demo, the metrics, or the win. Documenting them is a strength judges reward (it shows production-headroom awareness). Net infra remains **₹0** with comfortable room for the hackathon.

Resource	Free limit	Risk	Mitigation
Supabase DB	500 MB	measurement volume	retain recent + aggregate; archive to Storage/R2
Supabase project	**pauses after 7 days idle**	DB asleep at demo	keep-alive ping via Actions cron; unpause 24h before demo
Earth Engine	usage quotas	satellite job throttling	nightly precompute + cache features
Gemini API	rate/day limits	live demo throttling	cache advisories/dossiers; batch; Groq/Ollama fallback
Cloud Run	2M req/mo, cold start	demo lag	pre-warm; scale-to-zero off during pitch
GitHub Actions	2000 min/mo (private repo)	pipeline minutes	**use a public repo → unlimited free minutes** (+ open-source pitch point); else lean + stagger
Colab/Kaggle	session/GPU caps	training interrupts	checkpoint to Storage; small models first
Twilio (IVR)	trial credit; **trial calls only to verified numbers**	judge can't call cold	pre-verify the demo/judge number, or use in-app call sim; **Telegram bot is the free default**

23. Build Order (maps to PRD phases)

PRD Phase	Architecture deliverable	"Done" =
P0 Foundation	repo, Supabase schema, connectors (Delhi/Blr/Mum), H3, base map	live AQI on map, 3 cities
P1 Attribution	A1 + ML attribution + SHAP + blame map UI + validation vs inventory	blame map + ±15–20% agreement
P2 Forecast	A2 + LightGBM + persistence backtest + time-slider	skill ≥0.25 vs persistence
P3 Action	A3+A4, RAG corpus, dossiers, Telegram/IVR, i18n, latency traces	signal→action <5 min, live IVR
P4 Scale	A5 + `/admin/cities` live onboarding + comparative tab	4th city onboarded on stage
P4.5 AI upgrades	E1 CV detection, E2 dense-coverage, E3 what-if, **E5 optimiser, E6 multimodal evidence, E7 health/carbon**, E4 spike (stretch)	CV feeds enforcement; dense 1km renders; what-if re-forecasts; optimiser ranks packages; dossiers cite satellite patches; ₹/health/CO_2e on cards
P5 Polish	deploy, deck, demo video, `evaluate.ipynb`	internal dry-run scores 5/5

24. Open Technical Decisions

1. **Forecast finale model:** GNN vs Temporal Fusion Transformer — decide after MVP backtest (pick whichever beats persistence most on held-out data).
2. **Embeddings:** Gemini (free, hosted) vs local `bge-small` (zero-API) — default local to avoid rate limits.
3. **IVR for finale:** keep Twilio trial, or simulate in-app call UI to stay 100% free? (Recommend: Twilio trial for the "judge calls a real number" wow; in-app fallback ready.)
4. **Traffic data:** OSM + temporal proxy (free) vs a free-tier traffic API — start with proxy; note as a known limitation.
5. **Confirm** product name **VayuNetra** for repo + branding.

End of ARCHITECTURE.md v1.4 (synced to PRD v1.4). Next artifacts on request: (a) the data-source integration runbook (exact APIs, keys, EE scripts), (b) the validation/backtesting notebook plan, (c) the pitch-deck outline, (d) a P0 starter scaffold (repo + Supabase migrations + one working connector).